Adaptive Sampling Toward a Dynamic Graph Convolutional Network for Hyperspectral Image Classification

Yun Ding, Jinpeng Feng, Yanwen Chong, Shaoming Pan, and Xiaohui Sun

***Abstract*— Graph convolutional networks (GCNs) have been shown to be effective for hyperspectral image (HSI) classification due to their capacity to learn representations of spatial–spectral features. However, the existing GCN-based models heavily rely on predefined receptive fields to capture and aggregate neighbor information for each node, which limits the ability to adaptively selecting the most significant receptive field from graph data. To address the aforementioned problem, in this article, we propose a novel dynamic adaptive sampling GCN (DAS-GCN) algorithm that captures neighbor information through adaptive sampling to allow the receptive field to be dynamically obtained. The basic underlying idea is that the most meaningful receptive field for each target node can be adaptively discovered, and the edge adjacency weights can be adjusted simultaneously after each adaptive sampling operation. Thus, we enable the graph to be dynamically updated and refined. Specifically, the adaptive sampling operation consists of two complementary components; in the first step, the importance of different remote nodes in a large-scale neighborhood is learned, while in the second step, rich underlying spatial–spectral information is extracted from local neighbors and filtered. The proposed model has the ability to learn how to extensively exploit spectral–spatial correlations from both local and remote nodes. Moreover, the proposed DAS-GCN model has a superior ability to leverage node feature information to naturally generalize and efficiently generate node embeddings for unseen data. The experimental results with overall accuracy on four real HSI datasets, i.e., Indian Pines, Pavia university, Houston 2013, and Salinas are 95.63%, 96.40%, 94.70%, and 99.08%, respectively, which clearly demonstrate the advantages of the proposed method compared with other state-of-the-art approaches.**

***Index Terms*— Adaptive sampling, dynamic graph convolutional network (GCN), hyperspectral image (HSI) classification, receptive field, remote nodes, spatial–spectral.**

I. Introduction

HYPERSPECTRAL images (HSIs), which contain hundreds of narrow contiguous wavelength bands throughout the electromagnetic spectrum with high spectral resolution, can be easily acquired at present owing to the rapid development of hyperspectral sensors. Based on the inherently rich and detailed spectral signatures of HSIs [1]–[5], HSI classification has been successfully used in a variety of fields, such as urban mapping, vegetation monitoring, and mineral exploration.

To date, a variety of approaches have been successfully designed to address the HSI classification problem. Among them, early-stage approaches mainly focused on fully exploiting spectral information by adopting traditional machine learning methods, such as K-nearest neighbors [6] and support vector machines (SVMs) [7], which were typically trained on handcrafted features extracted from HSI data. However, the lack of consideration of the spatial correlations among pixels made it difficult to classify different land-cover categories accurately. To address this shortcoming, many researchers have attempted to incorporate spatial features into the classification process. These spectral–spatial classification methods include the Markov random field (MRF) approach [8], Gabor filtering [9], and hierarchical segmentation [10]. In particular, among these approaches, morphological profile-based methods [11], [12] have shown good performance in terms of classification accuracy because of their advantages in effectively combining spatial and spectral information.

However, the aforementioned approaches heavily rely on prior information empirically designed for HSI data, resulting in a limited discriminative ability on different datasets. Inspired by the achievements enabled by deep learning (DL) techniques in the fields of natural image classification, DL-based methods [13] have led to significant progress in the area of HSI classification because of their strong representation ability. The first attempt to use stacked autoencoder networks for HSI classification was designed by Chen *et al.* [14], in which low-level features were gradually aggregated to automatically learn and generate abstract high-level representations without complicated feature engineering. However, the aforementioned model concentrated only on the extraction of spectral features and failed to make full use of the correlations among different image pixels, resulting in large spatial variability of spectral signatures. Consequently, to alleviate this problem, Ma *et al.* [15] attempted to extract spectral–spatial features and achieved superior performance based on a DL framework via a supervised strategy. In addition, a recurrent neural network (RNN)-based architecture was first presented by Mou *et al.* [16] for application to HSI classification task.

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Subsequently, as an alternative to fully connected networks, which need more parameters for the same number of hidden layers, convolutional neural networks (CNNs) have drawn considerable attention because of their advantages in HSI classification. For instance, in [17] and [18], CNNs were used to capture the local stationary structures of spectral and spatial information through multiscale hierarchical patterns to achieve promising performance in HSI classification. In addition, considering that feature extraction and classifier training fail to share a uniform objective function in some traditional spectral–spatial classification architectures, a spectral–spatial unified network (SSUN) [19] was proposed for HSI classification, in which spectral and spatial features are simultaneously integrated. Moreover, to address the drawbacks of high dimensionality and restricted availability of training samples for hyperspectral data, Chen *et al.* [20] introduced a 3-D CNN-based feature extraction model in which L2 regularization is adopted to efficiently exploit deep spectral–spatial information and avoid the overfitting phenomenon. Furthermore, to significantly increase the number of available training samples to take full advantage of the capabilities of CNNs, a CNN with pixel-pair features (CNN-PPFs) [21] was presented, in which the pixel-pair features are learned by combining the center pixel with its neighborhood to enhance the distinguishing power in HSI classification. Wang *et al.* [22] proposed a graph convolutional broad network (GCBN) for semisupervised HSI classification and designed a combinatorial average method (CAM) for use in the model training phase to use valuable paired samples to generate a sample expansion set, thereby further improving the feature representation ability. In addition, other architectures from computer vision have provided new insights for the classification of HSI. For example, Song *et al.* [23] constructed a network layer by layer via residual learning [24] to extract more discriminative features. Zhou and Prasad [25] introduced a deep feature alignment neural network, which uses deep convolutional RNNs to extract and align discriminative features.

Despite the remarkable performance achieved by the existing CNN-based methods, they still suffer from some shortcomings. For example, when the available training samples are limited, conventional CNN-based classifiers have difficulty achieving promising classification performance because their large numbers of parameters may lead to network overfitting. Moreover, conventional supervised CNN models cannot make full use of the enormous number of unlabeled instances that can be easily collected to improve the distinction among different classes [26], [27]. In addition, the power of CNNs is still limited, especially when used on irregular regions in HSI patches due to the fixed shape of the convolution kernels. To be concrete, such inflexible convolutional kernels will inevitably fail to capture the intrinsic or extrinsic geometric relationships between different object regions in hyperspectral data.

To overcome this issue, graph convolutional networks (GCNs) [28] have emerged as a powerful tool for HSI classification tasks. By combining neighborhood structures with node features in their convolution operations, semisupervised graph-based methods can effectively aggregate feature information across the neighbors of each graph node to achieve satisfactory results. The existing GCN based methods can be broadly divided into two categories: spectral GCNs and spatial GCNs. For spectral GCNs, their objective is to learn a frequency response function to achieve spectral filter on given signal. For example, Bruna *et al.* [29] proposed the first spectral convolution neural network (spectral CNN) and developed the operation of graph convolution based on the spectral graph theory. Later, Defferrard *et al.* [30] extended the spectral CNN with fast localized convolution by defining a filter as Chebyshev polynomials of the diagonal matrix of eigenvalues. Kipf and Welling [28] formally proposed a GCN concept by simplifying the first-order approximation of spectral graph convolution, resulting in a more efficient filtering operation. Considering that the GCN only captures static sample structural information which lacks feature representativeness during the convolution process, Fu *et al.* [31] proposed a dynamic spectral graph learning convolutional network (DLGCN) to fit the local geometry information by optimizing the one-order Chebyshev polynomial. Next, considering that most GCNs only use the local geometry of row vectors of high-dimensional data via example graph Laplacian while neglecting the manifold structure information of column vectors, Fu *et al.* [32] proposed the example feature spectral GCN (EFGCN) to achieve effective semi-supervised classification, which simultaneously uses the example graph Laplacian and feature graph Laplacian to better preserve the local geometry distributions of data. For spatial GCNs, their objective is to find a node aggregation function to learn proper aggregation obtaining an updated node embedding. For example, Qin *et al.* [33] proposed the spectral–spatial GCN (S^2GCN) model for HSI classification, which simultaneously incorporates intrinsic spectral features and relative local spatial information from the receptive field. Different from the S^2GCN model, which operates on a fixed graph, Wan *et al.* [34] proposed the multiscale dynamic GCN (MDGCN) model to enable dynamic updating of the graph and to capture diverse spectral–spatial information from receptive fields at multiple scales. Furthermore, Jiang *et al*. [35] proposed the dynamic hypergraph neural network (DHGNN) to encode high-order data relationships from adjusted feature embedding while the DHGNN method cannot be scalable to large graph because of its full graph training. Based on the MDGCN model, considering that the receptive field of a GCN model is often limited to a fairly small region, Wan *et al*. [36] proposed the context-aware dynamic GCN (CAD-GCN) model to further discover long-range contextual relationships for HSI classification. In addition, for handling large graph, Hong *et al.* [37] developed a new minibatch GCN (FuNet), which combines the receptive fields of CNNs with those of GCNs to extract different types of HSI features using three fusion schemes. Hamilton *et al*. [38] proposed the minibatch-based graph sample and aggregate (GraphSAGE) methods within a general inductive framework to classify unseen samples.

Nevertheless, the aforementioned GCN-based methods have two common limitations. Specifically, the above-mentioned GCNs usually perform convolutions that are defined directly on the graph with a predesigned receptive field [39]. Moreover,

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the selected neighboring nodes are invariant after construction of the adjacency matrix, even though the graph edge weights may be dynamically refined throughout the graph convolution process [34], [36]. These GCN-based methods focus only on how to update the edge weights of pairwise pixels without considering the meaningfulness of the neighbors themselves. However, since hyperspectral data may be contaminated by noise, similarity measurements among the initial input pixels may not be accurate. Therefore, it is also worth considering that a predefined receptive field may not remain optimal throughout the entire convolution process in GCN-based methods. For instance, the process of updating the adjacency weights with poor sampled neighbors will have only a limited effect in boosting classification results. Consequently, the ability to efficiently extract spatial information from neighbors is also limited, ultimately weakening the expressive capabilities of the generated representations. Thus, a question arises: is there an adaptive and automated way of choosing the receptive field in the graph that contributes most effectively to representation throughout the whole network training process? In addition, most GCN-based architectures cannot provide a satisfactory solution for precisely generalizing to inference on unseen nodes, which still presents a potential challenge for the scalability of the GCNs applied for HSI classification in practice.

Inspired by the recent success of graph networks for node classification [40]–[42], previous authors have demonstrated a significant impact on graph representation learning by assigning attention coefficients or biases to sampled differentiated neighbors of a target node. In this article, we introduce a new type of GCN model called adaptive sampling toward a dynamic GCN (DAS-GCN) to address these difficulties and further boost the performance of HSI classification. Instead of using a predefined receptive field for convolution, we design an adaptive sampling operation, by which the most meaningful receptive field can be discovered for each target node in each layer and the related edge adjacency weights can be adjusted simultaneously. Essentially, the most meaningful receptive field for a target node in the graph domain is equivalent to a selected set of significant neighbors of the target node. Consequently, the update process after each adaptive sampling operation can be divided into two parts, namely, the refinement of the receptive field and the refinement of the corresponding connection relationships. As a result, the model is able to gradually produce discriminative embedded feature representations and a faithful graph. In addition, compared with FuNet, the proposed DAS-GCN not only is an inherently inductive method but also achieves better classification performance. Our adaptive sampling operation consists of two complementary components: in the first step, the importance of different distant nodes in a large-scale neighborhood is learned, while in the second step, rich underlying spatial–spectral information is extracted from the local neighbors of the previously selected nodes and filtered. The proposed model has the ability to learn how to extensively exploit spectral–spatial correlations from both local and remote neighbors, facilitating the generalization of our trained model and allowing it to leverage node feature information to efficiently generate promising node embeddings for new input samples. In summary, the major contributions of this article are as follows.

1) We propose a novel adaptive sampling operation, which can dynamically refine the receptive field of each target node and the corresponding connection relationships simultaneously throughout the whole network training phase.
2) The two complementary components of this operation, which are carried out successively in each round of adaptive sampling, work collaboratively to extensively exploit rich underlying spectral–spatial information from both local and remote neighbors, thus providing a satisfactory solution for generalization to precise inference on unseen nodes.
3) The experimental results demonstrate that the proposed model has a superior ability to precisely infer samples (including unseen data) and achieves remarkable classification performance compared with the existing classification methods.

The remainder of this article is organized as follows. After a brief introduction of background knowledge relevant to GCN models in Section II, we elaborate on the proposed DAS-GCN model in Section III. Section IV reports extensive experimental results and corresponding analyses. Finally, our conclusions are summarized in Section V.

II. Related Work

In this section, we first follow [39] in briefly reviewing the principles of the GCN model, including basic definitions and notations, the definition of a graph, and graph convolution. For clarity, each step is presented in detail as follows.

A. Definition of a Graph

Formally, let an undirected graph be defined as $G = (V, \varepsilon)$, where V is a series of nodes $\{\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_N\}$ and ε denotes the edge set. Each node $\boldsymbol{v}_i$ represents a b-dimensional spectral feature vector $\boldsymbol{x}_i \in \mathbb{R}^b$. The entire feature matrix $\mathbf{X} \in \mathbb{R}^{N\times b}$ consists of N feature vectors, $\mathbf{X} = [\boldsymbol{x}_1, \ldots, \boldsymbol{x}_N]^{\mathrm{T}}$.

The sparse symmetric adjacency matrix, denoted by $\mathbf{A} \in \mathbb{R}^{N\times N}$, indicates the connections between each pair of nodes. $\mathbf{A}_{ij}$ can be calculated using the following radial basis function (RBF):

$$\mathbf{A}_{ij} = \exp\left(-\frac{\left\|\boldsymbol{x}_i - \boldsymbol{x}_j\right\|^2}{\gamma}\right) \tag{1}$$

where the parameter γ is empirically set to control the width of the RBF. The vectors $\boldsymbol{x}_i$ and $\boldsymbol{x}_j$ represent the spectral signatures associated with the HSI pixels $\boldsymbol{v}_i$ and $\boldsymbol{v}_j$, respectively. The diagonal degree matrix is defined as $\mathbf{D} = \mathrm{diag}(d_1, \ldots, d_N)$, where $d_i = \sum_{j=1}^{N} \mathbf{A}_{ij}$ denotes the ith row sum of the adjacency matrix.

B. Graph Convolutions

Compared with the multilayer perceptron (MLP) framework [43], [44], the GCN model has a superior ability to efficiently aggregate features. Consider the following expression

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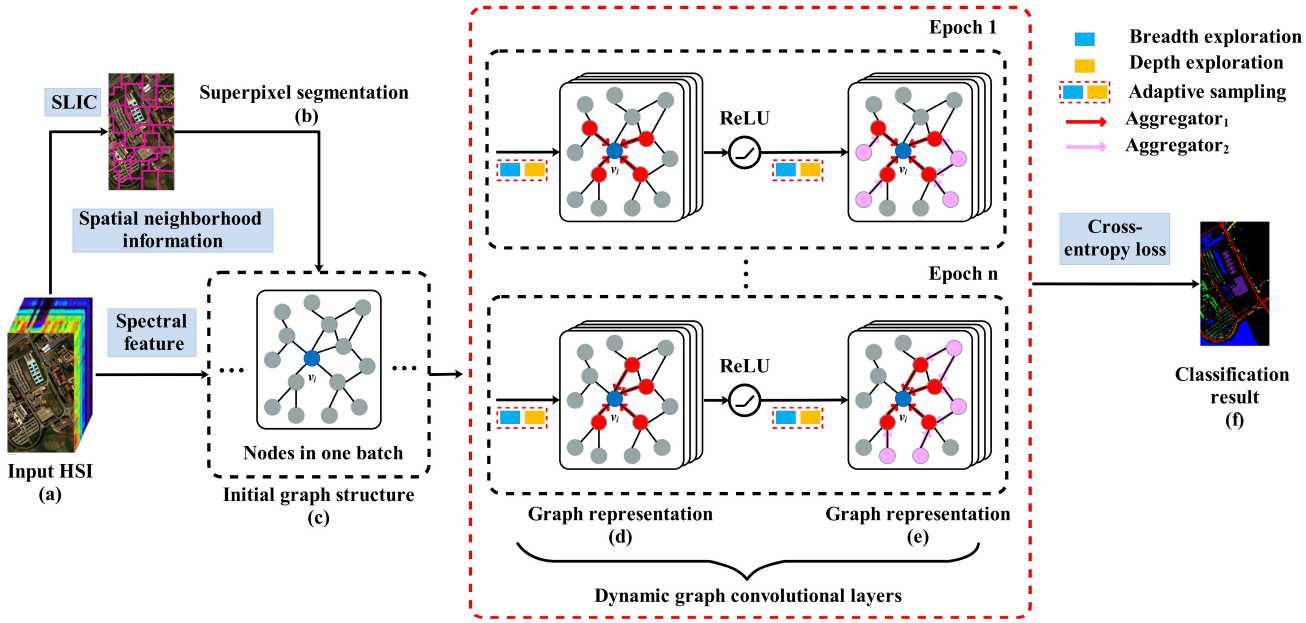


Fig. 1. Illustration of our proposed DAS-GCN model. (a) Original HSI. (b) Superpixels segmented by means of the SLIC algorithm [37], which help guide the construction of the initial edge set. (c) We use a minibatch approach to build an initial graph for one batch. The blue node represents the target node $\boldsymbol{v}_i$, and solid lines denote the adjacency relationships. (d) and (e) Two dynamic graph convolutional layers. From (d) to (e), the nodes represented in red and pink are the sampled neighbors of the target node $\boldsymbol{v}_i$ in each of the two layers, respectively, while the gray nodes are filtered out and are not sampled. As this convolution process is iterated, both the selected significant neighbors and the corresponding edge weights are dynamically refined after each adaptive sampling operation, and thus, discriminative embedded feature representations and a faithful graph can be generated. Specifically, our adaptive sampling operation consists of two complementary components designed for breadth and depth exploration of the receptive field. (f) Finally, the cross-entropy loss is used to penalize the label differences between the network output and all the labeled instances.

for feedforward propagation in typical GCNs developed by Kipf and Welling [28]:

$$\mathbf{h}^{(l+1)}(\boldsymbol{v}_i) = \sigma\left(\sum_{j=1}^{N} \mathbf{A}_{ij}\mathbf{h}^{(l)}(\boldsymbol{u}_j)\mathbf{W}^{(l)}\right), \quad i = 1, \ldots, N. \quad (2)$$

Here, $\mathbf{A} = \mathbf{D}^{-1/2}\mathbf{A}\mathbf{D}^{-1/2}$, where $\mathbf{A} = \mathbf{A} + \mathbf{I}$ and $\mathbf{D}_{ii} = \sum_{j=1}^{N} \mathbf{A}_{ij}$ are defined as the renormalization terms of $\mathbf{A}$ and $\mathbf{D}$, respectively. Kipf and Welling applied the renormalization trick with added self-loops to alleviate the problems of numerical instabilities and vanishing gradients in a deep neural network. $\sigma(\cdot)$ is a nonlinear activation function, and $\mathbf{W}^{(l)} \in \mathbb{R}^{D^{(l)} \times D^{(l+1)}}$ is the filter matrix in the lth layer. $\mathbf{h}^{(l+1)}(\boldsymbol{v}_i)$ denotes the hidden embedded feature in the $(l+1)$th layer for node $\boldsymbol{v}_i$, which aggregates intermediate feature information from the target node's sampled neighborhood in the lth-layer, $\boldsymbol{u}_j \in \mathcal{N}(\boldsymbol{v}_i)$.

As a result, the convolution operation of all the nodes over the entire graph can be expressed as follows:

$$\mathbf{H}^{(l+1)} = \sigma(\tilde{\mathbf{A}}\mathbf{H}^{(l)}\mathbf{W}^{(l)}) \quad (3)$$

where $\mathbf{H}^{(l+1)}$ and $\mathbf{H}^{(l)}$ denote the output and input feature representations of the $(l+1)$-th layer and the l-th layer, respectively. $\mathbf{H}^{(0)} = \mathbf{X}$, and $\mathbf{H}^{(0)}$ is the input feature representation of the first GCN layer.

The last layer of the GCN model, similar to the MLP model, predicts the labels of all the nodes using softmax classification. Let $\mathbf{Z} \in \mathbb{R}^{N \times C}$ be the network prediction results for N nodes, where $\mathbf{Z} = \text{soft}\max(\mathbf{H}^{(L)})$ and C denotes the number of classes. L is the number of graph convolutional layers, and $\mathbf{Y}$ contains all the labels for the labeled nodes. For semisupervised node classification in a multiclass task, the cross-entropy loss is commonly used to evaluate the performance of a model over all the labeled nodes, namely,

$$L_{\text{GCN}} = -\sum_{i \in y}\sum_{j=1}^{C} \mathbf{Y}_{ij} \ln \mathbf{Z}_{ij} \quad (4)$$

where y is the set of node indices with ground-truth labels.

III. Proposed Method

In this section, we detail the proposed DAS-GCN model for HSI classification. A schematic of the DAS-GCN model is presented in Fig. 1. Given an input image [Fig. 1(a)], we first obtain superpixels [Fig. 1(b)] segmented by means of the simple linear iterative clustering (SLIC) algorithm [45], which helps us guide the construction of the initial edge set. We use a minibatch approach to build an initial graph [Fig. 1(c)] for one batch. Then, dynamic graph convolution [Fig. 1(d) and (e)] is performed, in which both the selected significant neighbors and the corresponding edge weights can be dynamically refined in each adaptive sampling operation. Finally, the classification result [Fig. 1(f)] is obtained by the softmax classifier. Meanwhile, Fig. 2 presents the 2-D t-distributed stochastic neighbor embedding (t-SNE) [38] visualizations of the feature maps output by the two convolutional layers of the GCN and DAS-GCN models as well as visualization of the raw features

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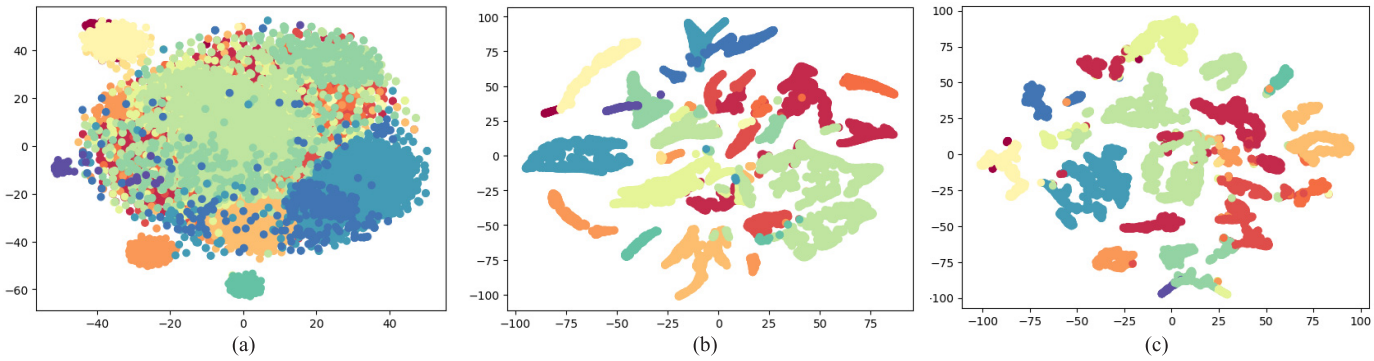


Fig. 2. 2-D t-SNE visualizations of the feature maps output by the two convolutional layers of the GCN and DAS-GCN models as well as raw features on the Indian Pines dataset. Different classes are marked in different colors. One can observe that the data of different classes are distributed more clearly in our DAS-GCN representation, which demonstrates the desired discriminative ability of DAS-GCN for semisupervised HSI classification tasks. (a) Raw features. (b) GCN features. (c) DAS-GCN features.

on the Indian Pines dataset. The critical operations in the proposed DAS-GCN model are detailed in this section.

A. Superpixel Segmentation

In fact, the graph structure information is not originally available in an HSI [36]. Therefore, we need to use the relationships between pairs of HSI pixels to construct the edge set. In this article, we use a segmentation algorithm named SLIC [45] to segment the entire input image into a small number of compact superpixels. Theoretically, each superpixel is a local homogeneous region that consists of many pixels with a strong spectral–spatial relationship. Unlike in conventional superpixel-based methods [36], each superpixel is treated as a graph node. We attempt to guide the construction of the edge set based on the superpixel map. We aim to build an initial edge between a pair of pixels $\boldsymbol{v}_i$ and $\boldsymbol{v}_j$ when both $\boldsymbol{v}_i$ and $\boldsymbol{v}_j$ lie in the same superpixel. In addition, the feature vectors of the graph nodes, $\boldsymbol{x}_i$ and $\boldsymbol{x}_j$, denote the spectral signatures associated with HSI pixels $\boldsymbol{v}_i$ and $\boldsymbol{v}_j$, respectively. Therefore, we can obtain an initial graph that is coarse but preliminarily reliable via the SLIC segmentation technique. The matrix $\mathbf{A}$ represents the dependencies of the adjacent nodes that lie in superpixels; its elements are defined as

$$\mathbf{A}_{ij} = \begin{cases} \exp\left(-\dfrac{D_{ij}}{\gamma}\right), & \text{if } \boldsymbol{v}_i, \boldsymbol{v}_j \text{ are in a superpiexel} \\ 0, & \text{otherwise} \end{cases} \tag{5}$$

where the parameter γ is empirically set to 6 in our experiments to control the width of the RBF. D_{ij} represents the spectral–spatial distance between pixels $\boldsymbol{v}_i$ and $\boldsymbol{v}_j$. Moreover, the spatial context can be regarded as weight values in the calculation of spectral similarity. Thus, we multiply the spectral distance (D_{spectral}) by the corresponding spatial distance (D_{spatial}), yielding the following expression:

$$D_{ij} = D_{\text{spectral}} \cdot D_{\text{spatial}}. \tag{6}$$

$x(v_i)$ and $x(v_j)$ are the spectral features of pixels $\boldsymbol{v}_i$ and $\boldsymbol{v}_j$, respectively; similarly, $c(v_i)$ and $c(v_j)$ are the spatial position coordinates of these pixels. The symbol $\cdot$ denotes the elementwise product operator. The spectral distance (D_{spectral}) between pixels $\boldsymbol{v}_i$ and $\boldsymbol{v}_j$ is defined as the spectral angle distance (SAD) measure

$$D_{\text{spectral}} = \cos^{-1}\left(\frac{\langle x(v_i), x(v_j)\rangle}{\|x(v_i)\|_2 \|x(v_j)\|_2}\right) \tag{7}$$

and the spatial distance (D_{spatial}) can be represented by the spatial Mahalanobis distance between pixels $\boldsymbol{v}_i$ and $\boldsymbol{v}_j$, as follows:

$$D_{\text{spatial}} = \sqrt{(c(v_i) - c(v_j))^{\mathrm{T}} M (c(v_i) - c(v_j))} \tag{8}$$

where M is a covariance matrix that represents the relationships and distribution of the structured graph nodes.

B. Adaptive Sampling

The major focus of our work is to attempt to learn a meaningful receptive field that can play a significant role in propagation and aggregation via our proposed adaptive sampling operation. Why is learning a meaningful receptive field for each node important to boost the performance of GCN-based HSI classification? The reasons are twofold: 1) the initial input graph may not be a faithful representation of the input image because the original HSI is often contaminated by noise and 2) different nodes represent different land-cover categories, possessing diverse spectral features and spatially related characteristics; thus, different receptive fields may be required for further exploration.

As shown in Fig. 3, our adaptive sampling operation consists of two complementary components: in the first step, the importance of different distant nodes in a large-scale neighborhood is learned, while in the second step, rich underlying spatial–spectral information aggregated from the local neighbors of the previously selected nodes is extracted and filtered. In the following, the critical underlying ideas of these two complementary components will be explained in detail in Sections III-C and III-D, respectively. In addition, to clarify the mathematical notations used in this article, some important notations are listed in Table I to clearly specify the relationships among variables and help readers understand the definitions of these mathematical notations.

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TABLE I
SUMMARY OF THE NOTATIONS

Notations	Descriptions
$\boldsymbol{v}_i$	the target node
V_t	a certain subgraph
$\boldsymbol{u}_j$	the initial neighbors of the target node
$\mathrm{N}_K(\boldsymbol{v}_i)$	the local spatial neighbors of the target node
$\mathrm{N}_M(\boldsymbol{v}_i)$	$\mathrm{N}_K(\boldsymbol{v}_i)$ and the target node itself
$\mathrm{N}_B(\boldsymbol{v}_i)$	all nodes that lie in the region explored during breadth exploration for the target node
$\bar{\boldsymbol{u}}_j : q(\boldsymbol{u}_j \mid \mathrm{N}_M(\boldsymbol{v}_i))$	node sampling for breadth exploration
$\mathrm{N}^{(V_t,l)}(\bar{\boldsymbol{u}}_j)$	sampled node set after breadth exploration of the receptive field for the target node from subgraph V_t in the l-th layer
$\bar{\boldsymbol{\mu}}_{js} \in \mathrm{N}_D(\bar{\boldsymbol{u}}_j)$	all nodes that lie in the region explored during depth exploration for the each previously selected node
$\mathrm{N}_C^{(V_t,l)}(\bar{\boldsymbol{\mu}}_{js})$	node set
d_{js}^{i}	the number of times each distinct node occurs in the node set
$\$\boldsymbol{u}_j : \%q(\bar{\boldsymbol{\mu}}_{js} \mid \mathrm{N}_M(\boldsymbol{v}_i))$	node sampling for the depth exploration
$\mathrm{N}_{adp}^{(V_t,l)}(\$\boldsymbol{u}_j)$	the meaningful neighbors of the target node
$\mathbf{A}_{ij}$	the initial adjacency matrix
$\mathbf{A}_{ij}^{V_t,l}$	the corresponding refined adjacency matrix for the target node from subgraph V_t in the l-th layer

C. Breadth Exploration of the Receptive Field

In the procedure described in this subsection, to explore the breadth of the receptive field for each target node, we assign different importance coefficients to one-hop neighbors at different distances in a large-scale neighborhood. The detailed algorithm process can be summarized as follows.

Given a node $\boldsymbol{v}_i$ from a certain subgraph V_t (i.e., $\boldsymbol{v}_i \in V_t$), in contrast to traditional nodewise sampling (i.e., $p(\boldsymbol{u}_j|\boldsymbol{v}_i)$), in which the selected neighbors (i.e., $\boldsymbol{u}_j$) are sampled only conditionally based on the parent $\boldsymbol{v}_i$, we develop a multinodewise importance sampling strategy for breadth exploration of the receptive field. Such sampling is efficient in two technical respects. First, the nodes we sample are visible and shared by the parent $\boldsymbol{v}_i$ with its local spatial neighbors $\mathcal{N}_K(\boldsymbol{v}_i)$. Specifically, the connections between the local spatial neighbors $\mathcal{N}_K(\boldsymbol{v}_i)$ and the target node $\boldsymbol{v}_i$ are captured and used to sample significant nodes, as shown in Fig. 3. Second, our breadth exploration process determines which neighbors are important to efficiently aggregate information for the target node $\boldsymbol{v}_i$. We denote the sampled nodes by $\boldsymbol{u}_j$ to distinguish them from the local spatial neighbors $\boldsymbol{v}_m \in \mathcal{N}_K(\boldsymbol{v}_i)$ and the target node $\boldsymbol{v}_i$. We use an attention mechanism [41], expressed as follows:

$$\alpha_{mj}^{i} = a(x(\boldsymbol{v}_m), x(\boldsymbol{u}_j)) = \frac{1}{n}\mathrm{ReLU}(W_1 x(\boldsymbol{v}_m), W_2 x(\boldsymbol{u}_j)). \quad (9)$$

Let $\alpha_{mj}^{i} = a(x(\boldsymbol{v}_m), x(\boldsymbol{u}_j))$ be the attention coefficient between the spectral feature of nodes $x(\boldsymbol{v}_m)$ and $x(\boldsymbol{u}_j)$, where

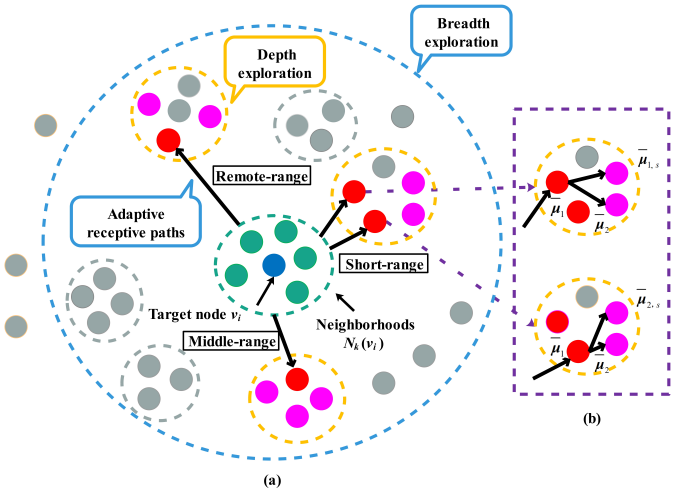


Fig. 3. Illustration of the implementation of breadth and depth exploration of the receptive field for the target node in the adaptive sampling at the first layer. In (a), the blue and green nodes represent the target node and its local spatial neighbors, respectively. We assume that the red nodes pointed by the black solid lines are selected nodes after breadth exploration, and the carmine nodes denote they are extracted after depth exploration, while the gray nodes imply that they are filtered out and unsampled. In (b), some nodes in the depth exploration of the target node may appear multiple times, which can be considered as frequency information. Different sizes of superpixels can be considered different sizes of neighborhoods, which can be adopted for breadth and depth exploration. Thus, we can learn how to explore the abundant spectral–spatial information and achieve better performance in terms of feature representation.

$\boldsymbol{v}_m \in \mathcal{N}_K(\boldsymbol{v}_i)$ represents an eight-connected neighbor of the target node $\boldsymbol{v}_i$ and $\boldsymbol{u}_j \in \mathcal{N}_B(\boldsymbol{v}_i)$ represents a node in a large-scale neighborhood of the target node $\boldsymbol{v}_i$. Considering that each superpixel consists of many pixels with a strong spectral–spatial relationship, the pixels within the superpixel in which the target node $\boldsymbol{v}_i$ is located can be regarded as the neighborhood of the target node $\boldsymbol{v}_i$. A large segmentation step (i.e., S_b) will result in a preference for large superpixels, which are equivalent to large-scale neighborhoods. W_1 and W_2 are the learnable parameters. $q(\boldsymbol{u}_j|\boldsymbol{v}_i \cup \boldsymbol{v}_1, \ldots, \boldsymbol{v}_m, \boldsymbol{v}_m \in \mathcal{N}_K(\boldsymbol{v}_i))$ is defined as the probability of sampling $\boldsymbol{u}_j$ among all the local spatial neighbors (i.e., $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_m, \boldsymbol{v}_m \in \mathcal{N}_K(\boldsymbol{v}_i)$) and the target node $\boldsymbol{v}_i$ itself. To simplify this expression, we adopt the notation $\mathcal{N}_M(\boldsymbol{v}_i)$ to represent $\boldsymbol{v}_i \cup \boldsymbol{v}_1, \ldots, \boldsymbol{v}_m, \boldsymbol{v}_m \in \mathcal{N}_K(\boldsymbol{v}_i)$. Therefore, the probability of sampling in this step is derived as follows: $q(\boldsymbol{u}_j|\mathcal{N}_M(\boldsymbol{v}_i)) = \alpha_{i,j}/(\sum_{m=1}^{\mathcal{N}_M(\boldsymbol{v}_i)} \sum_{j=1}^{\mathcal{N}_B(\boldsymbol{v}_i)} \alpha_{mj}^{i})$.

To avoid uncontrollable neighborhood expansion, similar to what is done in GraphSAGE [46], we uniformly sample a set of important neighbors with a fixed size (i.e., n_b) in this first step of the adaptive sampling operation. We define $\mathcal{N}^{(V_t,l)}(\overline{\boldsymbol{u}}_j)$, with $\overline{\boldsymbol{u}}_j \sim q(\boldsymbol{u}_j|\mathcal{N}_M(\boldsymbol{v}_i))$, as the sampled node set after breadth exploration of the receptive field for a target node from subgraph V_t in the lth layer. In this way, nodes that exhibit spectral similarity but are distant from the target node $\boldsymbol{v}_i$ can be captured since we adopt multinodewise importance sampling in a large-scale neighborhood.

D. Depth Exploration of the Receptive Field

The breadth exploration of the receptive field determines which neighbors (i.e., $\mathcal{N}^{(V_t,l)}(\overline{\boldsymbol{u}}_j)$) are important to the target

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node $\boldsymbol{v}_i$ and identifies the direction in which to explore rich underlying spatial–spectral characteristics. We will now describe how to perform depth exploration of the receptive field to extract and filter feature information from the local neighbors of the previously selected nodes [i.e., $\mathcal{N}^{(V_t,l)}(\overline{\boldsymbol{u}}_j)$] in this second step. We illustrate the idea of the two complementary components of the adaptive sampling operation in Fig. 3, which shows the implementation of breadth and depth exploration of the receptive field for the target node. The detailed algorithm process can be summarized as follows.

Given a previously selected node $\overline{\boldsymbol{u}}_j$, we define $\overline{\boldsymbol{\mu}}_{js} \in \mathcal{N}_D(\overline{\boldsymbol{u}}_j)$ as the local neighbors for depth exploration, specified by a small superpixel in which node $\overline{\boldsymbol{u}}_j$ is located. A small segmentation step (i.e., S_d) results in a preference for small superpixels. It is now commonly acknowledged that pixels in the same superpixel are spatially and spectrally related and that their labels are more likely to be the same [47], especially in a small superpixel. By collecting all the neighbors [i.e., $\overline{\boldsymbol{\mu}}_{js} \in \mathcal{N}_D(\overline{\boldsymbol{u}}_j)$] of each node in $\mathcal{N}^{(V_t,l)}(\overline{\boldsymbol{u}}_j)$, we obtain a node set $\mathcal{N}_C^{(V_t,l)}(\overline{\boldsymbol{\mu}}_{js})$. We observe that some nodes in $\mathcal{N}^{(V_t,l)}(\overline{\boldsymbol{u}}_j)$ may lie in the same superpixel, as shown in Fig. 3(b). For example, the node indices $\overline{\boldsymbol{\mu}}_{1,s}$ and $\overline{\boldsymbol{\mu}}_{2,s}$ may correspond to the same node when $\overline{\boldsymbol{\mu}}_1$ and $\overline{\boldsymbol{\mu}}_2$ lie in the same superpixel. Thus, we define d_{js}^i as the number of times that node $\overline{\boldsymbol{\mu}}_{js}$ occurs (or the frequency of node $\overline{\boldsymbol{\mu}}_{js}$) in the node set $\mathcal{N}_C^{(V_t,l)}(\overline{\boldsymbol{\mu}}_{js})$. A greater value of d_{js}^i indicates that node $\overline{\boldsymbol{\mu}}_{js}$ is more desirable for target node $\boldsymbol{v}_i$ and also much easier to select. We propose a logarithmic scale factor ρ_{js}^i to better describe this effect

$$\rho_{js}^i = \left(\log\left(d_{js}^i + 1\right)\right)^{\beta}, \quad d_{js}^i \geq 1,\ 0 < \beta < 1 \tag{10}$$

where β is a variable parameter that is negative for attenuation, positive for amplification, or 0 for no scaling

$$\overline{\alpha}_{is} = \lambda \times \exp(\alpha_{ij} \times A_{js}) + (1 - \lambda) \times \rho_{js}^i. \tag{11}$$

Here, α_{ij} is the attention coefficient between nodes $\boldsymbol{v}_i$ and $\overline{\boldsymbol{u}}_j$, as computed via (9). $\mathbf{A}_{js}$ denotes the adjacency value between nodes $\overline{\boldsymbol{u}}_j$ and $\overline{\boldsymbol{\mu}}_{js}$ as calculated via (5). $\lambda = 0.8$ is a weighting factor used to balance the terms of importance and frequency for depth exploration of the receptive field.

We define $\widetilde{q}(\overline{\boldsymbol{\mu}}_{js}|\mathcal{N}_M(\boldsymbol{v}_i)) = \overline{\alpha}_{is}/(\sum_{j=1}^{\mathcal{N}^{(V_t,l)}(\overline{\boldsymbol{u}}_j)} \sum_{s=1}^{\mathcal{N}_D(\overline{\boldsymbol{u}}_j)} \overline{\alpha}_{is})$ as the probability of sampling a node based on the previously selected nodes in this second step of the adaptive sampling operation. The final output of the sampled node set for a target node from a certain subgraph V_t in the lth layer is defined as $\mathcal{N}_{\text{adp}}^{(V_t,l)}(\hat{\boldsymbol{u}}_j)$, where $\hat{\boldsymbol{u}}_j \sim \widetilde{q}(\overline{\boldsymbol{\mu}}_{js}|\mathcal{N}_M(\boldsymbol{v}_i))$ and the size of $\mathcal{N}_{\text{adp}}^{(V_t,l)}(\hat{\boldsymbol{u}}_j)$ is fixed to n_d. We can define $\mathbf{A}_{ij}^{V_t,l}$ as the normalization of $\overline{\alpha}_{ij}$, where $\overline{\alpha}_{ij}$ denotes the refined adjacency value between the target node $\boldsymbol{v}_i$ and a sampled node $\hat{\boldsymbol{u}}_j$; the normalization equation is given as follows:

$$\mathbf{A}_{ij}^{V_t,l} = \alpha_{ij} = \overline{\alpha}_{ij} \Big/ \sum_{j=1}^{\mathcal{N}_{\text{adp}}^{(V_t,l)}(\hat{\boldsymbol{u}}_j)} \overline{\alpha}_{ij}. \tag{12}$$

Thus, the adjacency matrix $\mathbf{A}_{ij}^{V_t,l}$ can be dynamically updated based on the initial adjacency matrix $\mathbf{A}_{ij}$, where both the adjacent node indices and corresponding weight values are adjusted simultaneously. Accordingly, we perform the adaptive sampling operation to obtain the corresponding adjacency matrix $\mathbf{A}_{ij}^{V_t,l}$ for a target node $\boldsymbol{v}_i$ from a certain subgraph V_t in the lth layer.

E. Minibatch GCN

Because they suffer from heavy costs in terms of both computation and memory for neighborhood expansion across layers, traditional GCNs are not easily scalable to large graphs, especially for large-scale remote sensing datasets. To address this problem, inspired by inductive learning [48] and FuNet [37], we use a minibatch approach to improve the computational efficiency during training of embeddings. Note that [49] has theoretically guaranteed the applicability of the minibatch training strategy.

Instead of training individual embeddings for each node based on the full neighborhood, we design a function that allows the model to learn how to generate embeddings for each node in each batch by aggregating features from the corresponding meaningful neighborhoods, which inherently enhances the expressive power and enables the efficient generation of promising node embeddings for both the training set and new input samples. Note that some nodes from other batches may be sampled as meaningful neighbors for a target node in the current batch during the training phase. In general, however, the majority of the test nodes will be previously unseen to the model during training and thus can be treated as new input data for the proposed model.

Given an undirected graph $G = (V, \varepsilon)$ with $|V| = N$ as introduced in Section II-A, we randomly partition its nodes into a set of subgraphs: $\{G_t = (V_t, \varepsilon_t)|t = 1, \ldots, \lceil (N/M) \rceil\}$, where ε_t consists of the links for each node in V_t, including links between nodes in V_t and another subgraph, since pixels in the same superpixel may be randomly partitioned into different subgraphs. $\lceil \cdot \rceil$ denotes the ceiling operation, and M is defined as the number of nodes in a subgraph.

Given a node $\boldsymbol{v}_i$ from a certain subgraph V_t (i.e., $\boldsymbol{v}_i \in V_t$), we propose an adaptive sampling operation to select a fixed-size set (see Section IV-A) of significant neighbors $\mathcal{N}_{\text{adp}}^{(V_t,l)}(\widehat{\boldsymbol{u}}_j) \in \mathbb{R}^s$ of the target node $\boldsymbol{v}_i$ in the lth layer; then, $\mathbf{A}_{ij}^{V_t,l} \in \mathbb{R}^{1\times s}$ can be computed via (12) to indicate the relationship between $\boldsymbol{v}_i$ and an adjacent node $\widehat{\boldsymbol{u}}_j$ [i.e., $\widehat{\boldsymbol{u}}_j \in \mathcal{N}_{\text{adp}}^{(V_t,l)}(\widehat{\boldsymbol{u}}_j)$]. By collecting the adjacency matrix of each node $\boldsymbol{v}_i \in V_t$, the final adjacency matrix ($\mathbf{A}^{V_t,l} \in \mathbb{R}^{M\times S}$) of subgraph V_t (the batch) can be obtained through concatenation as follows:

$$\mathbf{A}^{V_t,l} = \begin{bmatrix} \mathbf{A}_1^{V_t,l} & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \mathbf{A}_M^{V_t,l} \end{bmatrix}. \tag{13}$$

Note that the dimensionality of the reassembled adjacency matrix is $M \times S$, where $S = M \times s$.

Once $\mathbf{A}^{V_t,l}$ is obtained, we apply the row normalization trick as follows:

$$\tilde{\mathbf{A}}_t^{(l)} = \mathbf{D}^{-1}\mathbf{A}^{V_t,l} \tag{14}$$

where $\mathbf{D}$ is a diagonal matrix representing the degrees of $\mathbf{A}^{V_t,l}$ and $\mathbf{A}^{V_t,l}$ is a nonsymmetric matrix. Using (3), the convolution

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operation of all the nodes over the subgraph can be directly expressed as

$$\mathbf{H}_t^{(l+1)} = \sigma\left(\tilde{\mathbf{A}}_t^{(l)}\mathbf{H}_t^{(l)}\mathbf{W}_t^{(l)}\right). \tag{15}$$

Here, t is not only the tth subgraph but also the tth batch for network training. $\tilde{\mathbf{A}}_t^{(l)}$ is the normalized and regularized adjacency matrix in the lth layer for the tth batch. $\mathbf{H}_t^{(l)}$ denotes the lth layer hidden embedding, $\mathbf{H}_t^{(0)}$ is the input feature representation of all the nodes in the tth batch, $\mathbf{W}_t^{(l)}$ represents the layer-specific parameters, σ denotes the activation function, and the final output for the full graph can be reformulated as

$$\mathbf{H}^{(l+1)} = \left[\mathbf{H}_1^{(l+1)}, \ldots, \mathbf{H}_t^{(l+1)}, \ldots, \mathbf{H}_{\lceil \frac{N}{M} \rceil}^{(l+1)}\right]. \tag{16}$$

F. Explicit Variance Reduction

The remaining question for the adaptive sampling procedure is how to define the exact form of the sampler $\widetilde{q}(\overline{\boldsymbol{\mu}}_{js}|\mathcal{N}_M(\boldsymbol{v}_i))$. Indeed, a good estimator should reduce the variance caused by the sampling process, since a high variance is likely to impede efficient training.

The variance of the spectral features for all the sampled neighbors of each node in subgraph V_t is given by

$$\boldsymbol{Var}_t = \frac{1}{M} \times \sum_{i=1}^{M}\left(\frac{\sum_{j=1}^{\mathcal{N}_{\text{adp}}^{(V_t,l)}(\widehat{\boldsymbol{u}}_j)} (\widehat{\alpha}_{ij} \cdot (x(\widehat{\boldsymbol{u}}_j) - x_{\text{mean}})^2)}{n_d}\right) \tag{17}$$

where $x_{\text{mean}} = (1/n_d)(\sum_{j=1}^{\mathcal{N}_{\text{adp}}^{(V_t,l)}(\widehat{\boldsymbol{u}}_j)} x(\widehat{\boldsymbol{u}}_j))$ and represents the mean value of the spectral features for all the sampled neighbors (i.e., $\mathcal{N}_{\text{adp}}^{(V_t,l)}(\widehat{\boldsymbol{u}}_j)$) of the target node $\boldsymbol{v}_i$. By taking classification loss [(4)] and variance [(17)] into account, we formulate a hybrid loss as follows:

$$L_{\text{GCN}} = -\sum_{i\in y}\sum_{j=1}^{C} \mathbf{Y}_{ij}^t \ln \mathbf{Z}_{ij}^t + \eta \boldsymbol{Var}_t. \tag{18}$$

Here, η is a trade-off parameter, which is fixed to 0.5 in our experiments. $\mathbf{Z}_{ij}^t$ represents the network prediction results for the nodes in subgraph V_t, while $\mathbf{Y}_{ij}^t$ represents the corresponding ground-truth labels. C denotes the number of classes, and y is the set of node indices with ground-truth labels. Note that for efficient computation, we penalize only the variance of the top layer in (18), and we find this sufficient to achieve promising performance in our experiments. Algorithm 1 summarizes our proposed DAS-GCN classification method.

Algorithm 1 DAS-GCN Model for HSI Classification

Input: Given the HSI data matrix, $\boldsymbol{X} = [\boldsymbol{x}_1, \ldots, \boldsymbol{x}_N] \in \mathbb{R}^{N\times b}$; segmentation steps, S_b and S_d; the total number nodes, N; the number of nodes in one batch, M; fixed sizes of the sampled node sets, n_b and n_d; and the number of graph convolutional layers, L;
1) Construct an initial full graph and the corresponding adjacency matrix via (1);
2) Partition the nodes of the initial graph into a set of subgraphs (batches);
3) **for** $t = 1$ to $\lceil (N/M) \rceil$ **do**
4) **for** $l = 1$ to L **do**
5) Obtain a sampled neighbor set for each node in the t-th subgraph via adaptive sampling;
6) Update and concatenate the corresponding $\mathbf{A}^{V_t,l}$ via (12) and (13);
7) Calculate the error term via (18), and update the weight matrices $\mathbf{W}_t^{(l)}$ using minibatch gradient descent;
8) **end for**
9) **end for**
10) Reformulate the output for the full graph via (16);
Output: Classification results for each pixel.

IV. Experimental Analysis

In this section, we present comprehensive experiments conducted to assess the effectiveness of the proposed DAS-GCN model on four real-world HSI datasets: Indian Pines, Pavia University (PaviaU), Houston University (Houston), and Salinas. We first consider other recent state-of-the-art methods for comparison, including GCN-based methods, namely, GCN [28], S^2GCN [33], MDGCN [34], CAD-GCN [36], DHGNN [35], and FuNet [37], and CNN-based methods such as CNN-PPF [21] and DPRN [50], as well as the traditional classification method known as joint collaborative representation and SVM with decision fusion (JSDF) [51]. Four evaluation metrics, namely, the per-class accuracy (CA), overall accuracy (OA), average accuracy (AA), and Kappa coefficient (KC) [2], are adopted to evaluate the performance of the proposed DAS-GCN model. Then, we analyze the influence of different training samples on the effectiveness of the proposed method in comparison to the other classification methods. After that, we demonstrate the performance benefits of the proposed model achieved via adaptive sampling design. Finally, we investigate the impact of the parameters in our proposed model.

A. Experimental Settings

In our experiments, the proposed DAS-GCN algorithm was implemented in PyTorch [52] with Adam optimizer [53]. The learning rate was set to 0.01, and two graph convolutional layers were adopted in the networks. The values of the segmentation steps S_b and S_d were set as described in Section IV-F. For the number of neighbors sampled in each layer, the values of n_b and n_d were set to $n_b = n_d = s = 15$ because a sample set size of 15 was found to provide a good balance between a reasonably short computation time and high feature representation capabilities. We set 30 training epochs per adaptive sampling operation to adjust the sampled neighbors and the corresponding adjacency weights. We usually randomly selected 30 labeled pixels (i.e., examples) per class for training; however, 15 labeled pixels were chosen if the corresponding category contained fewer than 30 examples. The rest of the pixels in each class were treated as unlabeled instances during the training process and were then used as the test set to assess the classification performance. The numbers of labeled and unlabeled pixels of different classes in the four datasets are listed in Table II. The batch size in the

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TABLE II
DETAILED NUMBER OF TRAINING AND TESTING SAMPLES ON THE INDIAN PINES, PAVIAU, AND HOUSTON DATASETS

	Indian Pines				PaviaU				Houston				Salinas			
Class	Name	train	test	Class	Name	train	test	Class	Name	train	test	Class	Name	train	test
1	Alfalfa	30	16	1	Asphalt	30	6601	1	Healthy grass	30	1221	1	Brocoli_green_weeds_1	30	1979
2	Corn-notill	30	13982	2	Meadows	30	186192		Stressed grass	30	1224	2	Brocoli_green_weeds_2	30	3696
3	Corn-mintill	30	800	3	Gravel	30	2069	3	Synthetic grass	30	667	3	Fallow	30	1946
4	Corn	30	207	4	Trees	30	3034	4	Tree	30	1214	4	Fallow_rough_plow	30	1364
5	Grass-pasture	30	453	5	Painted metal sheets	30	1345	5	Soil	30	1212	5	Fallow_smooth	30	2648
6	Grass-trees	30	700	6	Bare Soil	30	4999	6	Water	30	295	6	Stubble	30	3929
7	Grass-pasture-mowed	15	13	7	Bitumen	30	1300	7	Residential	30	1238	7	Celery	30	3549
8	Hay-windrowed	30	448	8	Self-Blocking Bricks	30	3652	8	Commercial	30	1214	8	Crapes_untrained	30	11241
9	Oats	15	5	9	Shadows	30	917	9	Road	30	1222	9	Soil_vinyard_develop	30	6173
10	Soybean-notill	30	942		Total	270	42506	10	Highway	30	1197	10	Corn_senesced_green_weeds	30	3248
11	Soybean-mintill	30	2425					11	Railway	30	1205	11	Lettuce_romaine_4wk	30	1038
12	Soybean-clean	30	563					12	Parking lot 1	30	1203	12	Lettuce_romaine_5wk	30	1897
13	Wheat	30	175					13	Parking lot 2	30	439	13	Lettuce_romaine_6wk	30	886
14	Woods	30	1235					14	Tennis court	30	398	14	Lettuce_romaine_7wk	30	1040
15	Buildings-Grass-Trees-Drives	30	356					15	Running track	30	630	15	Vinyard_untrained	30	7238
16	Stone-Steel-Towers	30	63						Total	450	14579	16	Vinyard_vertical_trellis	30	1777
	Total	450	9799										Total	450	53679

training phase was set to 512 to ensure that all the labeled pixels were contained in one batch, while in the test phase it was set to 8192 to control the number of batches. All the reported accuracies were computed based on the average over ten training sessions to obtain stable results.

B. Datasets Description

Four real-world benchmark HSI datasets, including Indian Pines, Pavia University, Houston University, and Salinas, are introduced in the following.

1) The Indian Pines image was acquired by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor during a flight campaign on June 12, 1992, over northwestern Indiana, USA. The scene consists of 145 × 145 pixels with 220 spectral channels. As a usual step, we retain 200 channels by removing 20 noisy and water absorption bands, i.e., 104–108, 150–163, and 220. Moreover, the spatial resolution was 20 m and the spectral resolution was 10 nm, covering a spectral range of 200 − 2400 nm. This scene covered an agricultural field and contained 16 main land-cover categories. Correspondingly, the false-color image and the ground truth are displayed in Fig. 4(a) and (b), respectively.
2) The Pavia University (PaviaU) dataset was gathered by the reflective optics system imaging spectrometer ROSIS) sensor during a flight campaign over Pavia University, Northern Italy, in 2002. The image comprises 610 × 340 pixels with 103 spectral channels ranging from 430 to 860 nm. Moreover, the geometric resolution of the sensor was 1.3 m, and nine land-cover types are included in the dataset. The false-color image and the corresponding ground-truth map of PaviaU are exhibited in Fig. 5(a) and (b), respectively.
3) The Houston University (Houston) dataset was captured by Compact Airborne Spectrographic Imager (CASI) over the campus of University of Houston and the surrounding urban area in June 2013, which has been used in the 2013 GRSS Data Fusion Contest. The Houston image contains 349 × 1905 pixels with 144 spectral channels ranging from 380 to 1050 nm. In addition, the spatial resolution was 2.5 m, and 15 challenging land-cover categories are included in the dataset. The false-color image and the corresponding ground-truth map of Houston are shown in Fig. 6(a) and (b), respectively.
4) Salinas is a classic large-scale HSI dataset which is collected by the AVIRIS sensor over Salinas Valley. This dataset comprises 204 spectral bands (20 water absorption bands are removed) and 512 × 217 pixels with high



Fig. 4. Indian Pines. (a) False-color image (bands 20, 69, and 178 are red, green, and blue, respectively) and (b) ground-truth image.

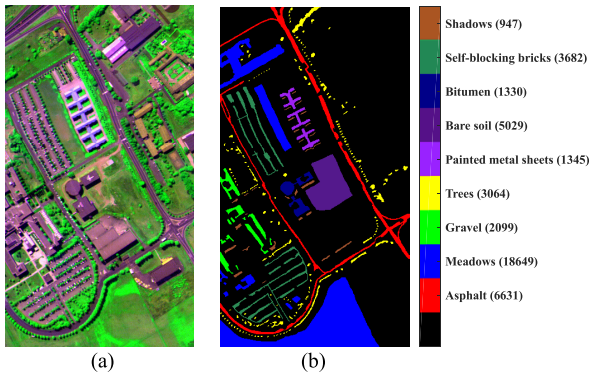


Fig. 5. PaviaU data. (a) False-color image (bands 52, 99, and 4 are red, green and blue, respectively) and (b) ground-truth image.

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TABLE III
OVERALL ACCURACY (STD) (%) OF DIFFERENT CLASSIFICATION METHODS ON THE INDIAN PINES DATASET

Class	GCN	S^2GCN	MDGCN	CADGCN	DPRN	CNN-PPF	JSDF	FuNet	DHGNN	DAS-GCN
1	95.00(2.80)	**100**(0)	**100**(0)	99.46(1.54)	100(0)	95.00(2.64)	**100**(0)	100(0)	83.33(14.91)	93.64(6.47)
2	56.71(4.42)	84.43(2.50)	80.18(0.84)	88.08(4.57)	74.6(8.51)	73.53(5.61)	90.75(3.19)	60.20(2.74)	40.89(3.53)	**94.67**(3.16)
3	51.50(2.56)	82.87(5.53)	**98.26**(0)	95.62(1.89)	85.42(5.54)	81.34(3.76)	77.84(3.81)	63.29(1.40)	48.55(8.88)	92.94(5.31)
4	84.64(3.16)	93.08(1.95)	98.57(0)	97.85(2.75)	99.12(0.83)	91.84(3.53)	**99.86**(0.33)	79.03(1.98)	65.99(4.40)	91.75(4.14)
5	83.71(3.20)	**97.13**(1.34)	95.14(0.33)	93.79(3.71)	93.34(4.26)	93.69(0.84)	87.20(2.73)	90.33(0.65)	81.64(4.62)	96.68(5.38)
6	94.03(2.11)	97.29(1.27)	97.16(0.57)	96.41(1.78)	97.24(2.88)	97.46(1.01)	98.54(0.28)	98.98(0.32)	87.51(4.32)	**98.60**(0.67)
7	92.31(0.00)	92.31(0)	**100**(0)	97.95(2.84)	**100**(0)	75.38(8.73)	**100**(0)	93.84(3.24)	83.33(7.57)	**100**(0)
8	96.61(1.86)	99.03(0.93)	98.89(0)	99.81(0.25)	99.18(0.91)	98.01(0.69)	99.80(0.31)	94.46(1.92)	85.69(5.09)	**100**(0)
9	**100**(0)	**100**(0)	**100**(0)	**100**(0)	**100**(0)	**100**(0)	**100**(0)	**100**(0)	76.67(23.38)	97.22(6.49)
10	77.47(1.24)	**93.77**(3.72)	90.02(1.02)	89.34(4.03)	85.53(4.61)	82.30(1.55)	89.99(4.24)	66.41(2.80)	67.01(4.21)	91.97(4.68)
11	56.56(1.53)	84.98(2.82)	93.35(1.47)	93.09(3.14)	75.34(4.17)	62.64(3.32)	76.75(5.12)	67.95(1.79)	64.49(3.39)	**98.49**(1.33)
12	58.29(6.58)	80.05(5.17)	93.05(2.30)	**93.41**(2.54)	79.04(4.96)	88.92(2.50)	87.10(2.82)	74.05(1.46)	45.51(6.39)	87.66(10.05)
13	**100**(0)	99.43(0)	**100**(0)	99.79(0.29)	99.91(0.23)	98.80(0.57)	99.89(0.36)	99.49(0.18)	93.54(2.88)	95.01(2.54)
14	80.03(3.93)	96.73(0.92)	**99.72**(0.05)	99.36(0.54)	92.92(3.55)	86.49(2.23)	97.21(2.78)	79.49(2.55)	87.46(3.18)	99.56(0.21)
15	69.55(6.66)	86.80(3.42)	**99.72**(0)	99.20(0.87)	96.07(3.04)	86.71(4.36)	99.58(0.68)	68.51(2.81)	44.99(7.54)	97.73(1.29)
16	98.41(0)	**100**(0)	95.71(0)	98.91(1.01)	99.74(0.65)	92.70(3.45)	**100**(0)	100(0)	98.43(0.77)	74.30(11.29)
OA	69.24(1.56)	89.49(1.08)	93.47(0.38)	94.13(0.78)	84.89(1.70)	80.09(1.56)	88.34(1.39)	73.71(0.47)	65.31(0.81)	**95.63**(0.92)
KC	65.27(1.80)	88.00(1.23)	92.55(0.43)	93.29(0.88)	82.86(1.91)	77.52(1.74)	86.80(1.55)	70.10(0.51)	72.19(2.11)	**95.01**(1.05)
AA	80.93(1.71)	92.99(1.04)	96.24(0.21)	96.38(0.35)	92.34(0.94)	88.57(0.77)	92.46(0.75)	78.59(0.37)	60.54(0.98)	**94.39**(1.12)
Time(s)	58	71	95	62	1794	1495	518	41	546	860

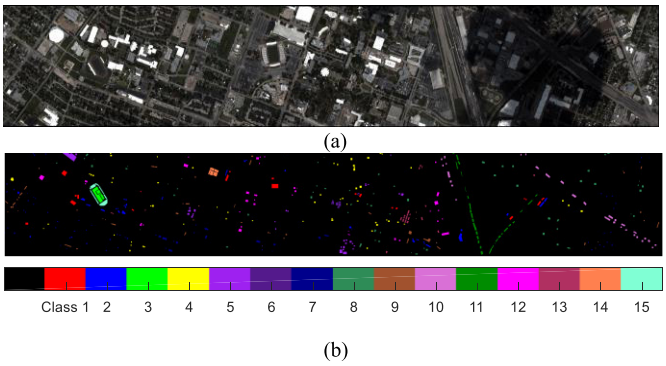


Fig. 6. Houston data. (a) False-color image (bands 45, 38, and 32 are red, green and blue, respectively) and (b) ground-truth image.

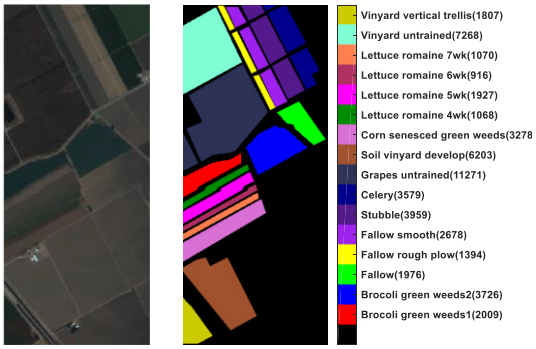


Fig. 7. Salinas data. (a) False-color image (bands 36, 17, and 11 are red, green and blue, respectively) and (b) ground-truth image.

spatial resolution (3.7-m pixels). The Salinas dataset contains 16 land-cover classes. The false-color image and the corresponding ground-truth map of Houston are shown in Fig. 7(a) and (b), respectively.

C. Experimental Results

Remarkably, our DAS-GCN yields stable performance improvements when compared with either DL-based methods or traditional methods. In particular, our model has a considerable classification advantage for large land-cover classes. Furthermore, the high classification scores demonstrate that our trained model is capable of better generalization to achieve precise inference for unseen nodes compared with the miniGCN-based model FuNet. The quantitative results obtained with the aforementioned methods on the four datasets are displayed in Tables III–VI, and the best value for each class (i.e., in each row) is highlighted in bold. The visual classification maps obtained with the different approaches on the various datasets are shown in Figs. 8–11. Classical HSI classification methods, such as JSDF, and DL-based methods (CNN-PPF and DPRN) have advantages in HSI classification compared with the GCN method, which proves the importance of considering spatial context. In addition, the classification performances of the S^2GCN, MDGCN, and CAD-GCN methods are generally better than those of the four aforementioned methods, which confirms the effectiveness of graph neural networks (GNNs) in terms of learning representations of spatial–spectral features.

1) *Results on the Indian Pines Dataset*: From Table III, it can be seen that the proposed DAS-GCN method achieves the best performance among all the methods in terms of OA and KC, reaching 95.63% and 95.01%, respectively. More specifically, a comparison with the other methods reveals that DAS-GCN shows superior classification performance on the "Corn-notill" and "Soybean-mintill" classes. However, our model yields poor classification accuracies on the "Stone-Steel-Towers" class.
2) *Results on the University of Pavia Dataset*: Similar to the results on the Indian Pines dataset, Table IV indicates that our proposed DAS-GCN method outperforms its competitors in terms of three quantitative criteria, namely, OA, AA, and KC. Compared with the

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TABLE IV

OVERALL ACCURACY (STD) (%) OF DIFFERENT CLASSIFICATION METHODS ON THE PAVIAU DATASET

Class	GCN	S^2GCN	MDGCN	CADGCN	DPRN	CNN-PPF	JSDF	FuNet	DHGNN	DAS-GCN
1	69.78(4.71)	92.87(3.79)	93.55(0.37)	83.85(4.80)	92.83(1.29)	95.73(0.80)	82.40(4.07)	85.26(0.90)	70.38(9.04)	**97.65**(1.01)
2	54.10(10.54)	87.06(4.47)	**99.25**(0.23)	95.56(1.97)	95.16(2.29)	84.01(1.99)	90.76(3.74)	97.61(0.49)	80.70(4.40)	99.15(0.22)
3	69.69(4.48)	87.97(4.77)	92.03(0.24)	**95.50**(2.41)	87.36(6.47)	86.45(1.94)	86.71(4.14)	86.25(0.73)	68.57(8.36)	86.36(2.31)
4	91.23(7.02)	90.85(0.94)	83.78(1.55)	81.90(3.51)	94.12(3.66)	91.70(2.06)	92.88(2.16)	98.51(0.55)	92.01(4.37)	**99.01**(0.68)
5	98.74(0.11)	**100**(0)	99.47(0.09)	98.91(1.06)	99.70(0.40)	99.93(0.04)	**100**(0)	99.87(0.12)	99.42(0.28)	98.12(0.97)
6	65.34(10.53)	88.69(2.64)	95.26(0.50)	**97.81**(2.89)	82.63(26.14)	93.57(1.28)	94.30(4.55)	84.26(1.34)	50.74(9.49)	94.66(2.81)
7	86.64(4.68)	98.88(1.08)	**98.92**(1.04)	96.79(2.42)	87.48(6.96)	93.53(0.72)	96.62(1.37)	83.51(1.05)	84.53(4.40)	93.38(4.35)
8	72.26(2.63)	89.97(3.28)	94.99(1.33)	**95.35**(2.76)	93.44(1.13)	83.83(1.60)	94.69(3.74)	71.04(2.00)	75.52(6.71)	89.07(3.41)
9	**99.93**(0.06)	98.89(0.53)	81.03(0.49)	84.80(5.33)	97.99(1.62)	99.47(0.34)	**99.56**(0.36)	100(0)	97.29(2.72)	91.51(3.53)
OA	66.19(3.43)	89.74(1.70)	95.68(0.22)	92.91(1.01)	92.69(2.36)	88.72(0.95)	90.82(1.30)	87.59(0.36)	76.40(2.14)	**96.40**(0.71)
KC	58.39(3.28)	86.65(2.06)	94.25(0.29)	90.69(1.29)	90.25(3.35)	85.43(1.18)	88.02(1.62)	85.46(0.42)	79.90(1.11)	**95.23**(0.93)
AA	78.63(1.23)	92.80(0.47)	93.15(0.28)	92.27(0.94)	92.30(3.24)	92.02(0.37)	93.10(0.65)	80.63(0.18)	69.27(2.66)	**94.39**(0.99)
Time(s)	1783	1803	244	73	3669	1545	1123	82	2340	392

TABLE V

CLASSIFICATION RESULTS (STD) (%) OF DIFFERENT METHODS ON THE HOUSTON UNIVERSITY DATASET

Class	GCN	S^2GCN	MDGCN	CADGCN	DPRN	CNN-PPF	JSDF	FuNet	DHGNN	DAS-GCN
1	88.16(1.90)	96.30(3.07)	93.42(4.25)	94.45(3.49)	97.53(1.45)	**98.62**(0.71)	97.41(1.21)	95.84(1.79)	95.66(1.87)	96.25(2.98)
2	97.20(0.48)	98.57(1.47)	93.67(3.60)	96.43(2.83)	97.83(1.40)	98.15(0.53)	**99.48**(0.25)	98.85(0.67)	97.93(0.95)	98.52(1.17)
3	97.91(0.13)	98.88(0.43)	98.12(1.09)	95.17(4.11)	99.91(0.13)	99.01(0.33)	99.88(0.22)	99.61(0.41)	98.63(0.73)	**99.98**(0.05)
4	96.55(0.41)	97.68(2.89)	95.58(1.85)	94.82(2.38)	94.93(1.46)	93.21(0.48)	**98.22**(2.80)	95.47(2.57)	96.20(1.14)	97.78(0.97)
5	89.79(0.71)	97.66(1.12)	99.00(1.30)	98.91(1.51)	99.40(0.96)	99.13(0.73)	**100**(0)	99.07(0.67)	93.34(6.18)	95.91(3.57)
6	98.21(1.15)	96.84(1.17)	93.28(6.08)	97.48(3.48)	95.19(3.50)	91.26(5.25)	99.32(1.09)	97.69(1.55)	97.43(0.66)	**99.52**(0.16)
7	73.67(1.94)	83.48(5.89)	87.68(4.41)	91.58(3.16)	88.56(1.89)	81.54(5.84)	91.93(4.91)	72.16(4.83)	70.60(17.88)	**93.92**(4.11)
8	65.71(4.64)	76.15(4.37)	80.45(6.12)	74.63(4.82)	79.64(4.72)	68.15(3.97)	68.82(6.16)	81.06(1.93)	62.97(6.97)	**95.57**(4.18)
9	70.27(3.03)	82.17(1.78)	89.64(2.26)	86.75(3.58)	81.51(4.93)	77.17(1.65)	69.47(8.56)	79.92(6.82)	16.02(36.46)	**93.20**(2.39)
10	74.71(2.32)	86.85(8.32)	90.06(6.41)	**94.23**(3.34)	93.40(2.91)	92.12(1.76)	85.63(9.32)	85.61(1.00)	27.27(3.39)	90.32(1.58)
11	75.36(2.37)	88.57(5.06)	86.73(3.22)	**94.65**(2.73)	89.86(4.59)	81.05(3.31)	94.51(3.82)	72.79(2.11)	56.60(39.19)	90.91(3.69)
12	79.29(4.80)	78.64(4.79)	89.44(5.69)	89.55(1.93)	**92.58**(4.40)	78.10(5.07)	84.33(5.33)	79.04(2.53)	55.10(29.33)	88.32(5.23)
13	12.09(2.68)	75.62(6.93)	92.78(4.45)	96.80(3.68)	96.95(1.57)	72.55(4.36)	**98.10**(1.28)	87.86(1.99)	39.39(11.85)	93.90(2.95)
14	86.03(3.31)	99.45(0.44)	99.43(0.97)	**100(0)**	**100(0)**	99.85(0.16)	100(0)	98.09(1.19)	95.75(1.13)	99.45(1.02)
15	95.29(1.67)	98.03(1.07)	96.27(1.72)	98.02(1.42)	99.84(0.23)	98.60(0.34)	99.86(0.36)	99.97(0.07)	97.80(0.80)	**100**(0)
OA	80.35(0.61)	89.31(1.00)	91.40(0.92)	92.51(0.73)	92.73(0.65)	87.54(1.03)	90.51(0.95)	87.07(0.61)	70.53(3.73)	**94.70**(0.79)
KC	78.72(0.66)	88.44(1.08)	90.70(1.00)	91.89(0.78)	92.14(0.70)	86.53(1.12)	89.74(1.03)	86.01(0.66)	73.38(3.17)	**94.27**(0.86)
AA	80.02(0.46)	90.33(1.06)	92.37(0.89)	93.57(0.60)	93.64(0.61)	88.57(0.77)	92.46(0.75)	88.32(0.36)	68.12(4.03)	**95.60**(0.53)
Time(s)	901	971	360	159	1637	1223	621	53	770	620

TABLE VI

OVERALL ACCURACY (STD) (%) OF DIFFERENT CLASSIFICATION METHODS ON THE SALINAS DATASET

Class	GCN	S^2GCN	MDGCN	CADGCN	DPRN	CNN-PPF	JSDF	FuNet	DHGNN	DAS-GCN
1	98.62(0.86)	99.01(0.44)	99.98(0.03)	100(0)	99.17(0.98)	99.77(0.21)	100(0)	99.27(0.32)	98.26(0.68)	100(0)
2	99.07(1.21)	99.18(0.59)	99.90(0.28)	100(0)	99.97(0.05)	98.69(0.89)	100(0)	99.94(0.03)	97.53(1.54)	100(0)
3	97.03(1.10)	97.15(2.76)	99.80(0.21)	99.97(0.07)	99.75(0.34)	99.50(0.49)	100(0)	96.13(1.99)	93.56(1.78)	99.98(0.03)
4	99.28(0.49)	99.11(0.55)	97.49(2.16)	98.46(1.48)	98.18(1.69)	99.81(0.04)	99.93(0.09)	99.85(0.06)	99.49(0.19)	97.93(0.47)
5	98.58(0.79)	97.55(2.35)	97.96(0.77)	97.77(1.05)	99.33(0.45)	96.64(1.26)	99.77(0.31)	96.04(4.52)	95.41(2.24)	99.72(0.34)
6	99.58(0.30)	99.32(0.35)	99.10(1.67)	99.61(0.33)	99.80(0.38)	99.32(0.86)	100(0)	99.90(0.07)	99.62(0.32)	100(0)
7	99.13(0.25)	99.06(0.27)	98.18(1.49)	99.10(1.56)	99.55(0.61)	99.59(0.13)	99.99(0.01)	99.61(0.20)	98.88(0.36)	100(0)
8	67.94(8.33)	70.68(5.20)	92.78(4.61)	95.80(3.17)	83.46(2.48)	74.77(4.01)	87.79(4.89)	69.57(7.02)	77.63(1.20)	99.53(0.36)
9	98.50(0.85)	98.32(1.79)	100(0)	**100(0)**	99.35(0.41)	**98.99(0.18)**	**99.67(0.33)**	99.39(0.31)	98.02(0.78)	99.98(0.01)
10	89.64(1.57)	90.97(2.59)	98.31(1.29)	98.01(1.40)	96.97(1.02)	89.32(3.04)	96.53(2.55)	91.82(2.02)	81.95(7.45)	99.43(0.36)
11	94.80(2.98)	98.00(1.65)	99.39(0.55)	99.82(0.27)	97.25(2.72)	97.65(1.49)	99.76(0.21)	96.22(2.37)	91.70(2.13)	100(0)
12	99.71(0.08)	99.56(0.59)	99.01(0.78)	97.99(0.78)	99.79(0.27)	99.82(0.30)	100(0)	99.63(0.74)	97.97(2.16)	99.99(0.02)
13	97.99(0.61)	97.83(0.72)	97.59(1.32)	97.90(0.98)	99.91(0.20)	97.70(0.50)	100(0)	99.47(0.78)	95.61(3.82)	97.67(3.39)
14	93.58(2.60)	95.75(1.65)	97.92(1.72)	98.89(1.53)	98.75(1.08)	94.14(1.22)	98.71(0.72)	96.63(2.52)	92.48(5.86)	96.98(2.20)
15	66.18(9.08)	70.36(3.62)	95.71(4.57)	97.47(1.65)	85.90(3.29)	79.12(1.99)	81.86(5.26)	62.52(3.77)	56.62(2.29)	95.64(2.99)
16	97.24(1.21)	96.90(1.97)	98.18(2.92)	99.76(0.57)	98.53(0.86)	98.65(0.31)	98.99(0.63)	98.56(0.43)	90.84(7.01)	100(0)
OA	87.16(0.85)	88.39(1.01)	97.25(0.87)	98.28(0.54)	94.07(0.53)	90.52(0.77)	94.67(0.77)	87.39(1.33)	86.59(0.23)	**99.08(0.44)**
KC	93.55(0.39)	94.30(0.47)	98.21(0.30)	98.78(0.22)	97.23(0.39)	95.22(0.34)	97.69(0.34)	88.50(0.57)	91.60(0.68)	**99.18(0.22)**
AA	85.74(0.92)	87.10(1.12)	96.94(0.96)	98.09(0.60)	93.41(0.60)	89.46(0.85)	94.06(0.85)	85.98(1.46)	85.04(0.26)	**98.97(0.49)**
Time(s)	3497	3528	1108	826	877	1769	3572	823	4278	3410

other methods, the classification results, particularly on the "Asphalt" and "Trees" classes, are superior by a significant margin, again confirming the power of our method.

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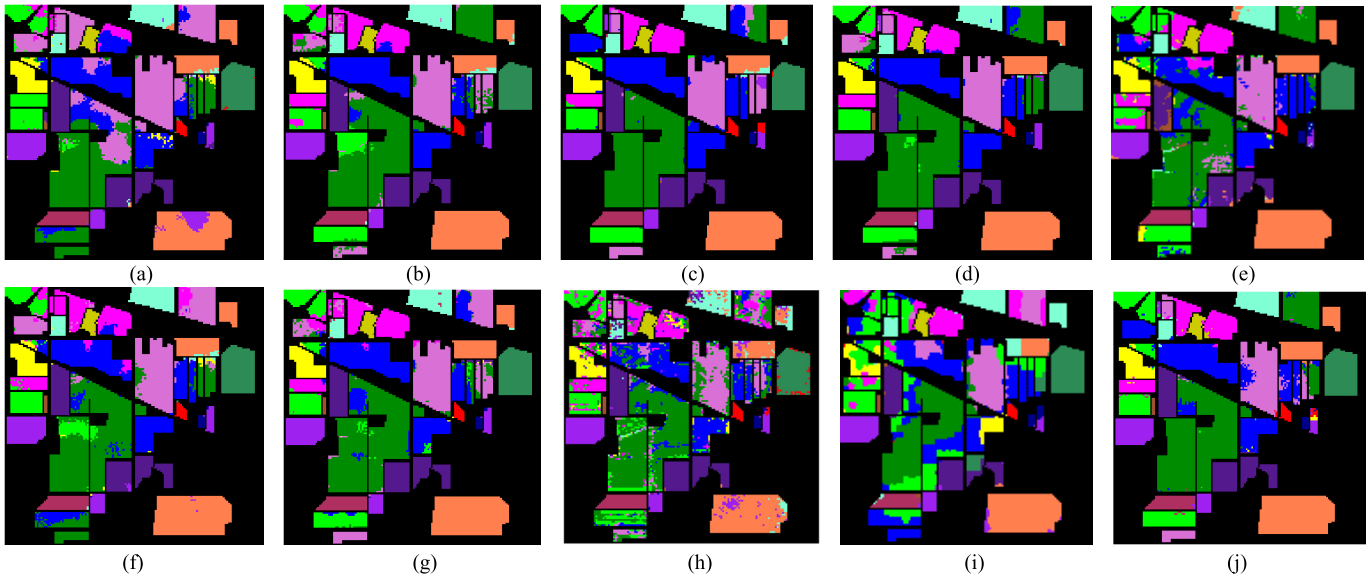


Fig. 8. Classification maps of different methods on the Indian Pines dataset. (a) GCN. (b) S^2GCN. (c) MDGCN. (d) CAD-GCN. (e) DPRN. (f) CNN-PPF. (g) JSDF. (h) FuNet. (i) DHGNN. (j) DAS-GCN.

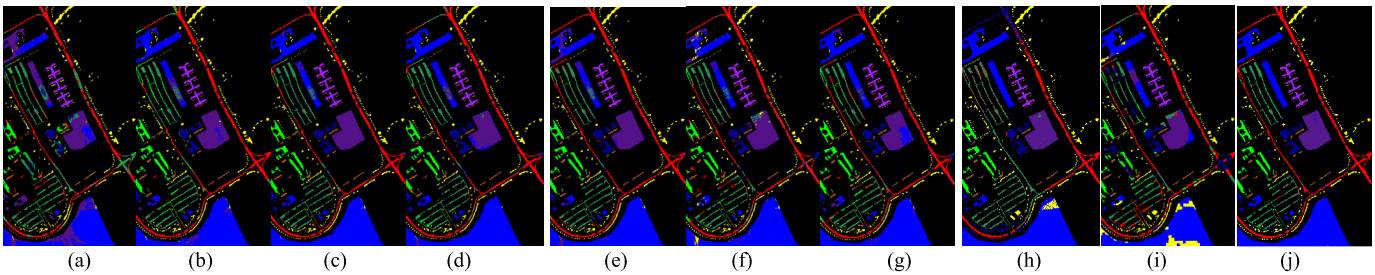


Fig. 9. Classification maps of different methods on the PaviaU dataset. (a) GCN. (b) S^2GCN. (c) MDGCN. (d) CAD-GCN. (e) DPRN. (f) CNN-PPF. (g) JSDF. (h) FuNet. (i) DHGNN. (j) DAS-GCN.

3) *Results on the Houston University Dataset*: From Table V, we can conclude that the proposed DAS-GCN method obviously outperforms the compared methods by a substantial margin. For instance, our model yields classification accuracies approximately 20% higher than those of all the other competitors on the "Commercial" class, which reveals the effectiveness of the adaptive sampling operation designed for our model.
4) *Results on the Salinas Dataset*: From Table VI, it can be seen that the proposed DAS-GCN method achieves the best performance among all the methods in terms of OA and KC, reaching 99.08% and 99.18%, respectively. Importantly, the classification performance of the proposed DAS-GCN method is very close to the ground truth compared with the other GCN-based methods to a certain degree.

D. Analysis of the Influence of Different Training Samples

To further investigate the classification performance of our DAS-GCN model, we varied the number of training samples per class from 5 to 30 in increments of 5 and recorded the specific OAs achieved by the aforementioned eight classification methods on the four datasets. In Fig. 12, it can be clearly observed that our proposed DAS-GCN model consistently outperforms the other classification methods with different numbers of labeled examples on the Indian Pines and Houston University datasets. We can observe from Fig. 10 that the classification results of our model improve rapidly with an increasing number of labeled instances on the University of Pavia dataset and that our model achieves better classification accuracies than MDGCN when the number of training samples reaches 15. All these observations support the advantageous performance and good stability of our DAS-GCN method.

E. Analysis of Model Depth

In this section, the classification results of the DAS-GCN method with different convolutional layers are evaluated on four datasets, which can be seen in Fig. 13. From Fig. 13, it can be seen that the DAS-GCN method is sensitive to the number of convolutional layers, and its performance decreases with the increasing layers. The main reason for this phenomenon is that high-order irrelevant features from different classes are aggregated into the core node to blur the discrimination of interclass information. Consequently, the two-layer DAS-GCN method is sufficient to capture intraclass information to achieve effective intraclass feature smoothing.

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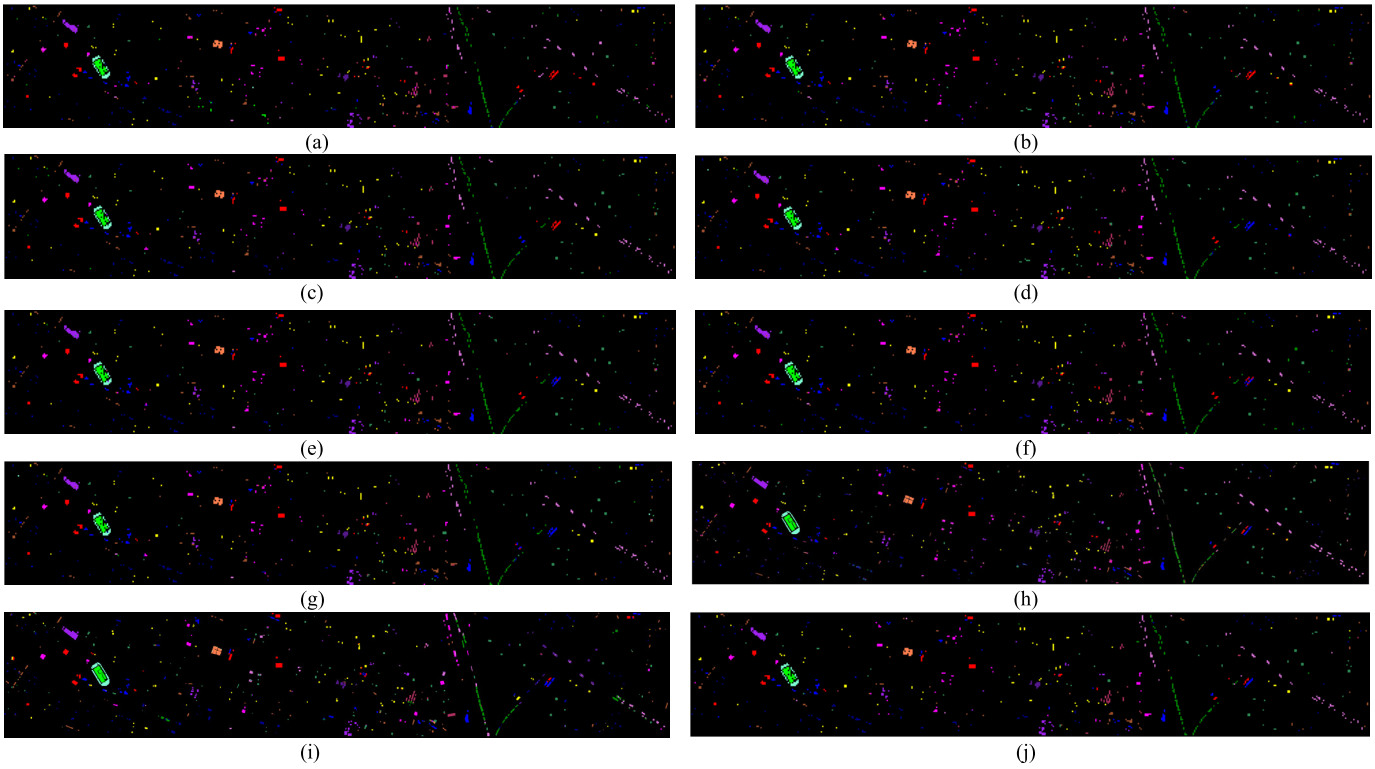


Fig. 10. Classification maps of different methods on the Houston dataset. (a) GCN. (b) S^2GCN. (c) MDGCN. (d) CAD-GCN. (e) DPRN. (f) CNN-PPF. (g) JSDF. (h) FuNet. (i) DHGNN. (j) DAS-GCN.

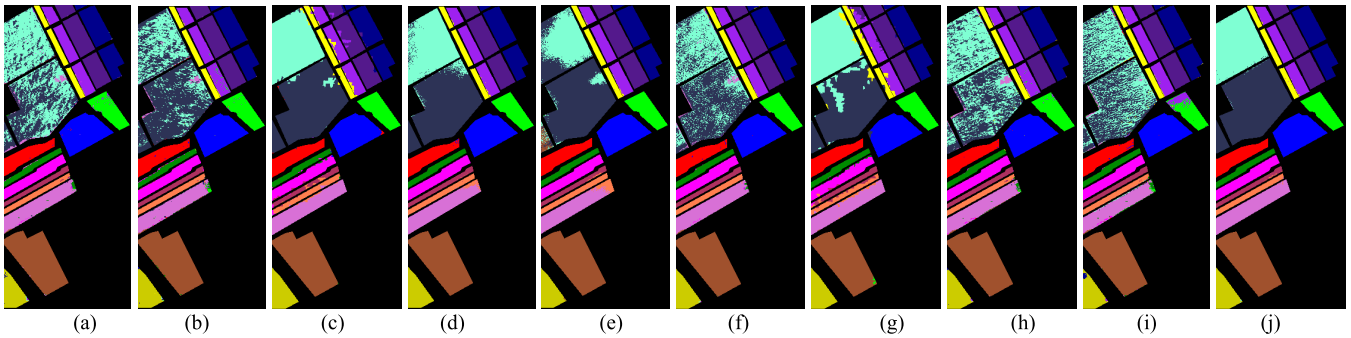


Fig. 11. Classification maps of different methods on the Salinas dataset. (a) GCN. (b) S^2GCN. (c) MDGCN. (d) CAD-GCN. (e) DPRN. (f) CNN-PPF. (g) JSDF. (h) FuNet. (i) DHGNN. (j) DAS-GCN.

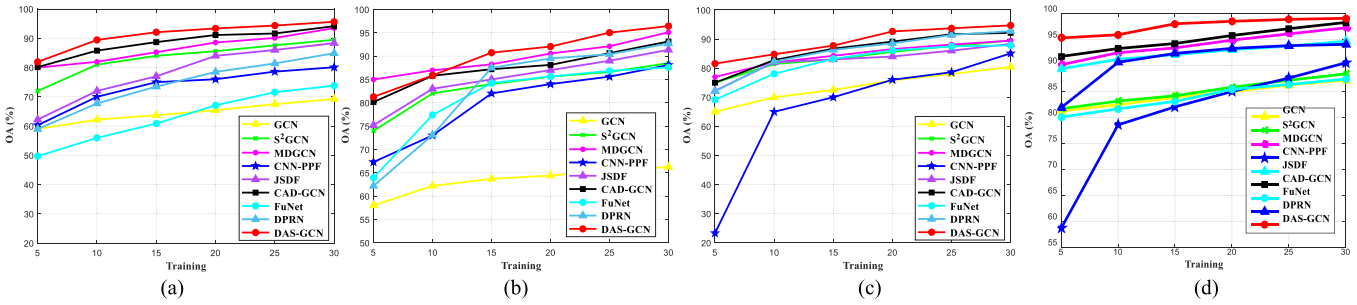


Fig. 12. Classification performance of the DAS-GCN model with different training samples. (a) Indian Pines. (b) PaviaU. (c) Houston. (d) Salinas.

F. Ablation Study

As mentioned in Section III, our proposed DAS-GCN architecture includes one critical component for improving the classification performance, that is, the adaptive sampling operation. In addition, we add a variance reduction in the training phase to significantly speed up convergence.

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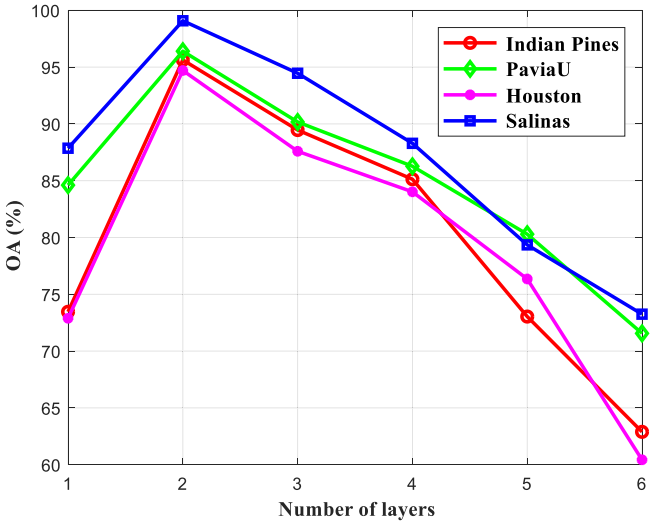


Fig. 13. Classification performance of different convolutional layers.

TABLE VII
CLASSIFICATION RESULTS (STD) (%) OF DIFFERENT METHODS ON THE INDIAN PINES DATASET

Class	DAS-GCN-r1	DAS-GCN-r2	DAS-GCN-r3	DAS-GCN-r4	DAS-GCN
1	93.24(12.12)	84.53(16.68)	78.64(26.66)	**94.54**(6.35)	93.64(6.47)
2	92.47(4.96)	**95.23**(2.17)	94.80(3.01)	94.39(3.23)	94.67(3.16)
3	90.39(5.80)	93.36(6.30)	92.38(6.02)	**93.87**(4.22)	92.94(5.31)
4	75.06(10.31)	85.71(7.75)	87.88(7.49)	90.69(4.72)	**91.75**(4.14)
5	98.55(1.13)	94.42(6.03)	**98.34**(1.75)	95.98(5.76)	96.68(5.38)
6	97.70(1.11)	97.82(1.13)	97.99(0.97)	98.53(0.78)	**98.60**(0.67)
7	**100**(0)	**100**(0)	97.44(5.02)	**100**(0)	**100**(0)
8	**100**(0)	**100**(0)	**100**(0)	**100**(0)	**100**(0)
9	**98.33**(5.27)	95.83(7.71)	93.75(8.63)	96.67(7.03)	97.22(6.49)
10	89.63(4.14)	89.75(5.31)	91.53(3.50)	89.18(5.09)	**91.97**(4.68)
11	98.49(1.10)	98.16(1.19)	98.76(0.69)	**98.60**(1.39)	98.49(1.33)
12	81.17(8.72)	**91.34**(3.75)	86.23(6.96)	89.32(8.44)	87.66(10.05)
13	93.99(3.20)	93.11(2.58)	94.98(2.05)	**95.93**(1.82)	95.01(2.54)
14	99.63(0.28)	**99.84**(0.15)	99.67(0.23)	99.57(0.21)	99.56(0.21)
15	95.72(2.09)	93.65(7.82)	96.35(2.33)	97.49(1.64)	**97.73**(1.29)
16	69.73(13.15)	69.99(9.63)	67.92(12.30)	70.15(12.21)	**74.30**(11.29)
OA	93.74(1.37)	95.19(1.42)	95.35(1.02)	95.43(0.84)	**95.63**(0.92)
KC	92.87(1.55)	94.50(1.61)	94.29(1.16)	94.79(0.95)	**95.01**(1.05)
AA	92.13(1.36)	92.67(1.78)	92.29(1.56)	94.06(1.09)	**94.39**(1.12)

To demonstrate the contributions of these novel components, we present the classification results of DAS-GCN obtained without each one of its four components on the three datasets. We use "DAS-GCN-r1" to denote that we perform adaptive sampling only at the beginning of the training procedure, which means that we perform graph convolution on the fixed initial graph and neighbors. "DAS-GCN-r2," "DAS-GCN-r3," and "DAS-GCN-r4" similarly represent the reduced models obtained by removing the first step, the second step, and the variance reduction step of the adaptive sampling operation, respectively. Tables VII–IX present the results obtained in this way on the aforementioned datasets. First, it is worth noting that when any one of these novel components is lacking, this will inevitably limit the classification performance. Second, we find that our "DAS-GCN-r1" model still performs better than S^2GCN on the fixed initial graph in terms of OA, AA, and KC, which indicates that our model can extensively exploit the rich underlying spectral–spatial information throughout the classification procedure and yield stable performance improvements. Third, compared with the results of DAS-GCN, the adaptive sampling operation enables a stable gain in performance with respect to "DAS-GCN-r1" since both the selected significant neighbors and the corresponding edge weights can be dynamically refined throughout the classification process.

TABLE VIII
CLASSIFICATION RESULTS (STD) (%) OF DIFFERENT METHODS ON THE PAVIA UNIVERSITY DATASET

Class	DAS-GCN-r1	DAS-GCN-r2	DAS-GCN-r3	DAS-GCN-r4	DAS-GCN
1	95.91(3.49)	94.91(1.28)	97.32(1.49)	97.43(1.07)	**97.65**(1.01)
2	98.59(0.12)	99.12(0.44)	**99.47**(0.41)	99.13(0.29)	99.15(0.22)
3	74.44(6.92)	81.65(1.72)	**87.59**(3.22)	86.14(4.85)	86.36(2.31)
4	96.25(3.01)	99.30(1.09)	**99.81**(0.22)	98.01(3.65)	99.01(0.68)
5	97.65(2.16)	**98.47**(0.77)	98.45(0.85)	98.01(0.89)	98.12(0.97)
6	87.48(3.98)	88.25(9.03)	88.32(6.24)	93.76(2.75)	**94.66**(2.81)
7	88.50(7.05)	87.91(8.39)	**94.43**(0.81)	91.32(4.35)	93.38(4.35)
8	85.50(5.33)	87.76(3.98)	88.89(1.86)	88.29(1.97)	**89.07**(3.41)
9	89.38(3.78)	85.08(3.81)	88.50(2.81)	92.35(4.34)	**91.51**(3.53)
OA	93.25(1.41)	94.25(1.69)	95.63(0.89)	95.98(0.54)	**96.40**(0.71)
KC	91.13(1.83)	92.42(2.17)	94.23(1.15)	94.69(0.70)	**95.23**(0.93)
AA	90.43(2.16)	91.39(1.88)	93.64(0.98)	93.83(0.72)	**94.39**(0.99)

TABLE IX
CLASSIFICATION RESULTS (STD) (%) OF DIFFERENT METHODS ON THE HOUSTON UNIVERSITY DATASET

Class	DAS-GCN-r1	DAS-GCN-r2	DAS-GCN-r3	DAS-GCN-r4	DAS-GCN
1	96.49(5.36)	93.56(5.26)	**97.39**(3.69)	94.16(1.22)	96.25(2.98)
2	96.18(5.76)	**99.94**(0.05)	92.42(3.83)	98.32(0.48)	98.52(1.17)
3	99.85(0.29)	99.61(0.53)	99.43(0.40)	99.63(0.32)	**99.98**(0.05)
4	**98.48**(1.02)	97.99(1.82)	97.42(0.90)	97.67(0.68)	97.78(0.97)
5	97.02(3.35)	97.91(3.00)	**98.56**(1.81)	97.80(2.28)	95.91(3.57)
6	97.78(3.59)	96.15(4.33)	99.7(0.26)	99.37(0.55)	**99.52**(0.16)
7	93.02(3.84)	93.39(4.60)	92.64(5.42)	**94.60**(3.02)	93.92(4.11)
8	96.65(0.81)	97.95(1.78)	**98.54**(1.50)	94.35(6.46)	95.57(4.18)
9	89.89(0.81)	88.04(4.21)	89.09(8.73)	89.37(3.93)	**93.20**(2.39)
10	87.36(5.89)	86.18(7.07)	85.59(1.97)	**92.88**(1.42)	90.32(1.58)
11	92.29(2.19)	90.51(2.89)	**94.41**(3.81)	92.86(3.44)	90.91(3.69)
12	87.23(2.64)	85.21(5.46)	87.32(2.56)	86.41(5.38)	**88.32**(5.23)
13	88.03(2.79)	81.97(9.48)	93.56(6.77)	88.30(7.35)	**93.90**(2.95)
14	99.75(0.49)	**99.87**(0.23)	99.43(0.99)	99.82(0.14)	99.45(1.02)
15	98.89(2.20)	97.95(1.78)	98.97(1.78)	**100** (0)	**100**(0)
OA	93.87(0.58)	93.11(0.81)	93.97(1.01)	94.32(0.27)	**94.70**(0.79)
KC	93.37(0.63)	92.55(0.87)	93.43(1.09)	93.85(0.29)	**94.27**(0.86)
AA	94.59(0.27)	93.78(0.94)	95.02(1.06)	95.11(0.26)	**95.60**(0.53)

G. Parameter Analysis

The performance of DAS-GCN depends on the adaptive sampling operation, which consists of two complementary components, namely, the breadth and depth exploration of the receptive field based on different scales of superpixel segmentation. The size of the segmentation step determines the shape and size of the superpixels [45]. As described in [54], a superpixel in an HSI is regarded as a local homogeneous region consisting of a certain number of pixels with a strong spectral–spatial relationship. Generally, a segmentation step that is too small (oversegmentation) may not provide sufficient node features to generate good embeddings, while a segmentation step that is too large may result in inaccurate segmentation (undersegmentation) and poor representation 1–5. Therefore, it is essential to investigate the performance gains

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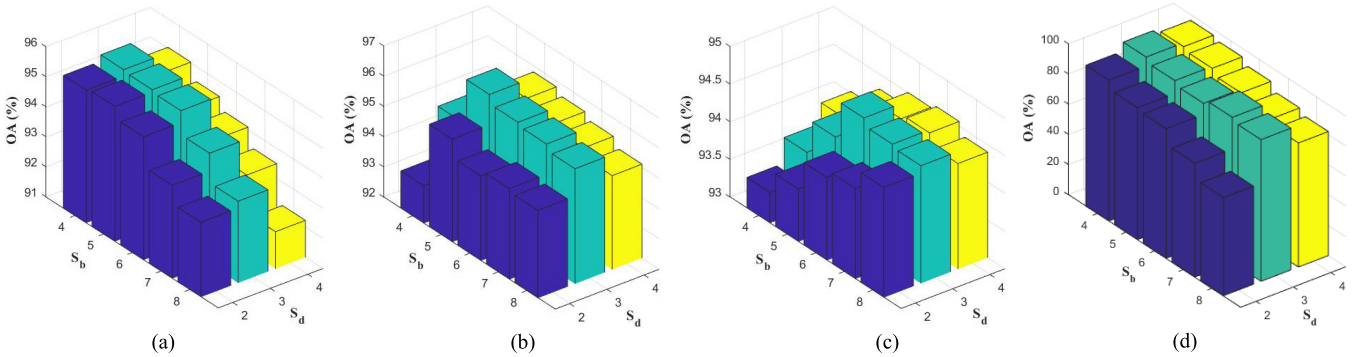


Fig. 14. Classification performance of the DAS-GCN model with different parameters. (a) Indian Pines. (b) PaviaU. (c) Houston. (d) Salinas.

of the proposed algorithm that can be achieved by manually tuning two parameters: the segmentation step for breadth exploration of the receptive field (S_b) and the segmentation step for depth exploration of the receptive field (S_d).

In Fig. 14, we evaluate in detail the sensitivity of the classification performance (in terms of OA) to different parameter settings of the DAS-GCN model on the four datasets. On one hand, we observe that the performance gain or degradation of the proposed method with the gradual variation in these two parameters is different on the four datasets. Most importantly, we find that setting the segmentation steps S_b and S_d to 6 and 3, respectively, leads to satisfactory performance for the Houston University dataset. However, this is not the case on the four datasets, for which the best classification performance is obtained when $S_b = 5$ and $S_d = 3$ for the former and $S_b = 4$ and $S_d = 3$ for the latter. Since SLIC segmentation is performed based on sampling on a regular grid, while the spatial structure distribution of the major land-cover categories in the University of Pavia dataset tends to be dominated by long shapes, a smaller segmentation step ($S_b = 5$) can yield more accurate superpixels to guide the breadth exploration of the receptive field. Considering that the sizes of some land-cover categories in the Indian Pines dataset are small, such as "Alfalfa," "Oats" and "Grass-pasture-mowed," a smaller segmentation step ($S_b = 4$) can generate more compact segments to guide the breadth exploration of the receptive field. Thus, it is necessary to carefully and comprehensively select a suitable combination of segmentation step parameter values for each dataset.

H. Complexity Analysis

In this section, we further analyze the complexity of the DAS-GCN algorithm via Algorithm 1. The main computational complexity is from feature information of the sampled neighbors via adaptive sampling technique. The complexity of feature information aggregation of the sampled neighbors is about $\boldsymbol{O}((N/M) \cdot M^2 \cdot Mb) = \boldsymbol{O}(NM^2b)$ in step 5 of Algorithm 1. The overall complexity of Algorithm 1 is about $\boldsymbol{O}(NM^2b)$ and $M \ll N$, $b \ll N$. Consequently, the DAS-GCN algorithm is suitable for large-scale HSI because the DAS-GCN method is almost linear with the number of total samples to a certain degree. The running time of the DAS-GCN method on four datasets can be seen in the bottom of Tables III–VI, which demonstrates that the running time of the DAS-GCN method is not the fast and not the lowest. The running time is reported on a server with NVIDIA 3090 GPU. In addition, with the improvements in computing capacity, the time problem will no longer be a major problem [55]. Importantly, the overall accuracy of the DAS-GCN method is the best compared with other classification methods in this article.

V. Conclusion

In this article, we present the novel DAS-GCN model for HSI classification task. Different from common GCN-based methods that work in the spatial domain by means of convolutions defined directly on the graph with predesigned receptive fields, the proposed DAS-GCN critically uses an adaptive sampling operation that allows the most meaningful receptive field to be discovered for each target node and the corresponding connection relationships to be simultaneously refined throughout the entire training phase. Meanwhile, the processes of breadth and depth exploration of the receptive field are carried out collaboratively to extensively exploit the rich underlying spectral–spatial information from both local and remote neighbors, facilitating generalization of our trained model and allowing it to leverage node feature information to more efficiently generate promising node embeddings for new input samples. Therefore, we can efficiently produce discriminative embedded feature representations and a faithful graph. Comprehensive experiments conducted on four real-world HSI datasets demonstrate the effectiveness and superiority of the proposed DAS-GCN model over its competitors in terms of OA, AA, and KC.

The proposed DAS-GCN method can still be improved in the future. For example, we plan to explore ways to adaptively select suitable values of the segmentation step parameters for different datasets to reduce the necessary effort of hand-designing parameters. In addition, it is still an interesting question that how to construct a deeper DAS-GCN method for effective HSI classification based on the inherent characteristic of HSI.

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